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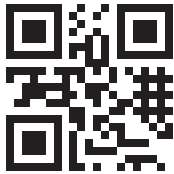
Financial Statement Analysis



SPOTLIGHT

Nestlé

www.nestlé.com



In 2016, Nestlé celebrated its 150th anniversary. From its humble beginning of making condensed milk, Nestlé now has over 2,000 brands under its umbrella, from coffee, snacks, baby food, and bottled water, to ice cream, pet care, and weight management products. Nestlé has assets of 132 billion Swiss Francs (CHF) and sales were just below CHF 90 billion. As a shareholder or potential investor, how would you evaluate Nestlé's financial performance and financial position? What analysis techniques can we use for financial analysis? How do we make sense of the numbers? ●

Jiradet ponari/123RF



The objective of financial reporting, through financial statements and other disclosures, is to help users make economic decisions. One of the primary tools available to you is a process we call financial statement analysis. This includes examining year-on-year changes, component and common-size analysis, as well as using the common financial ratios to help you make an economic decision on the financial position and financial performance of an entity.

In this chapter, we cover this process, using the financial statements of Nestlé Group, whose annual report is reproduced, in parts, in Appendix A. You can see Nestlé's financial statements summarized below.

A1				
	A	B	C	D
	Nestlé Group Consolidated Income Statement	For the year ended Dec. 31		
1				
2	(Adapted, in millions of Swiss Francs, CHF)	2016	2015	2014
3	Sales	89,469	88,785	91,612
4	Other revenue	317	298	253
5	Cost of goods sold	(44,199)	(44,730)	(47,553)
6	Distribution expenses	(8,059)	(7,899)	(8,217)
7	Marketing and administration expenses	(21,485)	(20,744)	(19,651)
8	Research and development costs	(1,736)	(1,678)	(1,628)
9	Other trading and operating income	453	204	264
10	Other trading and operating expenses	(1,597)	(1,828)	(4,175)
11	Operating profit	13,163	12,408	10,905
12	Financial income	121	101	135
13	Financial expense	(758)	(725)	(772)
14	Taxes	(4,413)	(3,305)	(3,367)
15	Income from associates and joint ventures	770	988	8,003
16	Profit for the year	8,883	9,467	14,904
17				

This chapter covers the basic tools of financial analysis. The first part of the chapter shows how we can evaluate Nestlé from year to year and how to compare it to other companies that are in the same industry. For this comparison, we can use other fast moving consumer goods (FMCG) companies around the world, including Procter and Gamble (P&G) and Unilever. The second part of the chapter discusses the more commonly used financial ratios. You have seen many of these ratios in the earlier chapters. However, we are yet to use all of them in a comprehensive analysis of a company.

By studying all these ratios together:

- You will learn the basic tools of financial analysis.
- You will enhance your ability to assess operations of a business.

Regardless of your chosen field—marketing, management, finance, entrepreneurship, or accounting—you will find these analytical tools useful as you move through your career.



A1				
	A	B	C	D
	Nestlé Group	At December 31		
	Consolidated Balance Sheet			
1	(Adapted, in millions of Swiss Francs, CHF)	2016	2015	2014
3	Cash and cash equivalents	7,990	4,884	7,448
4	Short-term investments	1,306	921	1,433
5	Inventories	8,401	8,153	9,172
6	Trade and other receivables	12,411	12,252	13,459
7	Prepayments and accrued income	573	583	565
8	All other current assets	1,361	2,641	1,884
9	Total current assets	32,042	29,434	33,961
10	Property, plant and equipment	27,554	26,576	28,421
11	Intangible assets and goodwill	53,404	52,008	54,357
12	Financial assets and other investments	16,428	14,094	14,142
13	All other non-current assets	2,473	1,880	2,569
14	Total non-current assets	99,859	94,558	99,489
15	Total assets	131,901	123,992	133,450
16				
17	Short-term borrowings	12,118	9,629	8,810
18	Trade and other payables	18,629	17,038	17,437
19	Accruals and deferred income	3,855	3,673	3,759
20	Provisions	620	564	695
21	All other current liabilities	2,295	2,417	2,194
22	Total current liabilities	37,517	33,321	32,895
23	Long-term borrowings	11,091	11,601	12,396
24	Provisions and other payables	5,027	4,330	5,003
25	All other non-current liabilities	12,285	10,754	11,272
26	Total non-current liabilities	28,403	26,685	28,671
27	Total liabilities	65,920	60,006	61,566
28	Share capital	311	319	322
29	Treasury shares	(990)	(7,489)	(3,918)
30	Retained earnings	82,870	88,014	90,981
31	Other equity items	(16,210)	(16,858)	(15,501)
32	Total equity	65,981	63,986	71,884
33	Total liabilities and equity	131,901	123,992	133,450
34				

A1				
	A	B	C	D
	Nestlé Group	For the year ended Dec. 31		
	Consolidated Cash Flow Statements			
1	(Adapted, in millions of Swiss Francs, CHF)	2016	2015	2014
3	Profit for the year	8,883	9,467	14,904
4	Adjustments (summarized)	6,699	4,835	(204)
5	Operating cash flows	15,582	14,302	14,700
6	Investing cash flows	(6,123)	(4,153)	(3,072)
7	Financing cash flows	(6,184)	(12,235)	(10,637)
8	Currency adjustments/retranslations	(169)	(478)	42
9	Net cash flows	3,106	(2,564)	1,033
10	Cash and cash equivalent at beginning of the year	4,884	7,448	6,415
11	Cash and cash equivalent at end of the year	7,990	4,884	7,448
12				

A1				
	A	B	C	D
1	Nestlé Group	For the year ended Dec. 31		
2	Consolidated Changes in Equity			
3	(Adapted, in millions of Swiss Francs, CHF)	2016	2015	2014
4	Equity at beginning of the year	63,986	71,884	64,139
5	Profit for the year	8,883	9,467	14,904
6	Other comprehensive income	741	(3,938)	1,900
7	Dividend paid	(7,369)	(7,374)	(7,219)
8	Other equity transactions	(260)	(6,053)	(1,840)
9	Equity at end of the year	65,981	63,986	71,884

LEARNING OBJECTIVES

1 Perform basic (horizontal and vertical) analysis of financial statements

2 Prepare common-size financial statements

3 Perform financial ratio analysis to make business decisions

4 Use financial ratios and other information to make investment decisions

It Starts with the Big Picture

Financial analysis involves more than just doing the math. Thorough analysis of the financial position and results of operations of a company begins with understanding the business and industry of the company—the big picture. This usually entails quite a bit of reading and research, using all kinds of media—the business press, trade journals, and other publications. You can often gain free access to this information via popular finance websites that you can research on the Internet, such as Google Finance, Yahoo Finance, MSN Finance, Bloomberg, Reuters, and many others.

There are also some “for pay” websites (such as the Motley fool, <http://www.fool.com>, and Dun & Bradstreet’s Hoovers, <http://www.hoovers.com>) or subscription services (such as Standard and Poor’s Capital IQ, www.capitaliq.com) through which industry and company analyses may be purchased. Learning about what’s happening in the industry, markets, general economic conditions, trends in product development, and specific company strategies puts the numbers in context and helps you understand why they turned out as they did. After all, accounting data should paint a picture of the results of implementing a particular business strategy. Once we have an understanding of the big picture, we can start to dig deeper into the numbers, and they will make more sense.

It is also impossible to evaluate a company effectively by examining only one year’s numerical data. This is why (in Chapter 4), we noted that *IAS 1—Presentation of Financial Statements* requires all financial statements to have comparative figures of at least two periods.

Nestlé’s financial statements for the last three financial years (2014–16) were shown at the beginning of this chapter. In fact, most financial analysis covers trends of three to five years. Since one of the goals of financial analysis is to predict the future performance of an entity, it makes sense to start by mapping the trends of the past. You saw an example of this with Singtel’s cash flow trends in Chapter 11. This is particularly true of Income Statement data such as net sales and net income.

Let’s start our examination of Nestlé with two basic analysis techniques: horizontal analysis and vertical analysis.

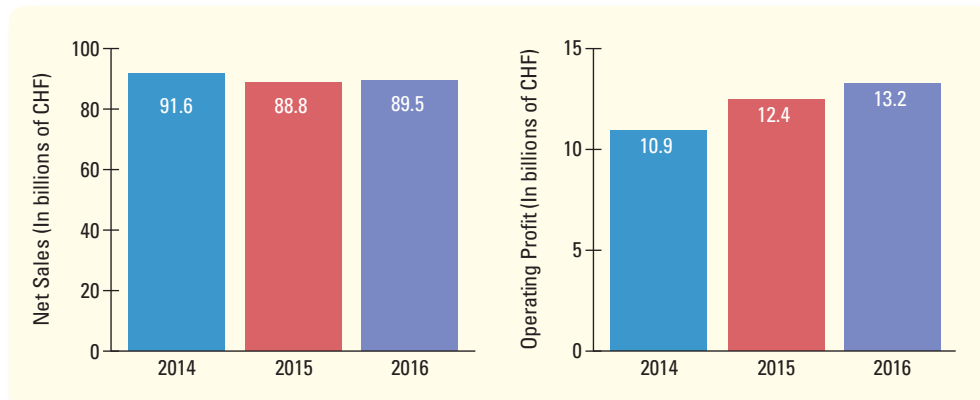
PERFORM BASIC (HORIZONTAL AND VERTICAL) ANALYSIS ON FINANCIAL STATEMENTS

1 Perform basic (horizontal and vertical) analysis of financial statements

Horizontal Analysis

Many decisions hinge on the trend of revenues, expenses, income from operations, and so on. Have revenues increased from last year? By how much? It is a good idea to start with a broad perspective of how Nestlé has performed in the last 3 years. We can simply look at the absolute numbers for sales and operating profit (or total assets or other totals), or we can show it visually. The graphs in Exhibit 12-1 show Nestlé's three-year trend of sales revenue and operating profit.

Exhibit 12-1 Sales and Operating Profit of Nestlé



It looks like Nestlé's sales have dropped somewhat, but it was able to increase its operating profit!

Suppose you are looking at another company and its sales have increased by \$50,000. In isolation, this fact is not very helpful, but knowing the long-term percentage change in net sales helps a lot. It is better to know that net sales have increased by 20% than to know that the increase is \$50,000. It's even better to know that percentage changes in net sales for the past several years have been rising year on year.

The study of percentage changes from year to year is called **horizontal analysis**. Computing a percentage change takes two steps:

1. Compute the amount of the change from one period (the base period) to the next.
2. Divide the amount of change by the base period amount.

Let's apply this to Nestlé's sales and operating profit for the year 2016 and 2015 (CHF 89,469 compared to CHF 88,785, and CHF 13,163 compared to CHF 12,408, respectively). For sales, the amount change is an increase of CHF 684, and for operating profit, an increase of CHF 755. Expressed as percentages, they represent an increase of 0.77% in sales and an increase of 6.08% in operating profit. As you can see, the percentages give a better context than just the actual sales or actual operating profits.

	2016	2015	Increase/(Decrease)	
			Amount	Percentage
Sales	89,469	88,785	684	0.77%
Operating profit	13,163	12,408	755	6.08%

If we extend this horizontal analysis to Nestlé's Income Statement, you will see something like this below in Exhibit 12-2.

Exhibit 12-2 Horizontal Analysis of Nestlé's Income Statements

A1					
	A	B	C	D	E
1	Nestlé Group Consolidated Income Statement	For the year ended Dec. 31		Increase/(Decrease)	
2	(Adapted, in millions of Swiss Francs, CHF)	2016	2015	Amount	Percentage
3	Sales	89,469	88,785	684	0.77%
4	Other revenue	317	298	19	6.38%
5	Cost of goods sold	(44,199)	(44,730)	(531)	(1.19%)
6	Distribution expenses	(8,059)	(7,899)	160	2.03%
7	Marketing and administration expenses	(21,485)	(20,744)	741	3.57%
8	Research and development costs	(1,736)	(1,678)	58	3.46%
9	Other trading and operating income	453	204	249	122.06%
10	Other trading and operating expenses	(1,597)	(1,828)	(231)	(12.64%)
11	Operating profit	13,163	12,408	755	6.08%
12	Financial income	121	101	20	19.80%
13	Financial expense	(758)	(725)	33	4.55%
14	Taxes	(4,413)	(3,305)	1,108	33.52%
15	Income from associates and joint ventures	770	988	(218)	(22.06%)
16	Profit for the year	8,883	9,467	(584)	(6.17%)
17					

Horizontal analysis does not provide you with answers as to why other income increases by 122.06% and other expenses decreases by 12.64%. You will need to carefully study the notes to the financial statements and make an assessment if these 2016 amounts are likely to repeat in the years beyond. You would want to check if there are likely to be changes in the income from associates and joint ventures and also the reasons why the tax expenses vary so much from 2015 to 2016. As an investor, you would want to assess if these items are likely to have a further impact in future operations. Horizontal analysis provides that first step in seeing how the numbers move from one year to the next.

Similarly, studying changes in Balance Sheet accounts can also enhance our understanding of the current and long-term financial position of the entity. Exhibit 12-3 looks at a few Balance Sheet changes that transpired for Nestlé in 2016.

Exhibit 12-3 Horizontal Analysis of Nestlé's Balance Sheets

A1					
	A	B	C	D	E
1	Nestlé Group Consolidated Balance Sheet	At December 31		Increase/(Decrease)	
2	(Adapted, in millions of Swiss Francs, CHF)	2016	2015	Amount	Percentage
3	Cash and cash equivalents	7,990	4,884	3,106	63.60%
4	Short-term investments	1,306	921	385	41.80%
5	Inventories	8,401	8,153	248	3.04%
6	Trade and other receivables	12,411	12,252	159	1.30%
7	Prepayments and accrued income	573	583	(10)	(1.72%)
8	All other current assets	1,361	2,641	(1,280)	(48.47%)
9	Total current assets	32,042	29,434	2,608	8.86%
10	Property, plant and equipment	27,554	26,576	978	3.68%
11	Intangible assets and goodwill	53,404	52,008	1,396	2.68%
12	Financial assets and other investments	16,428	14,094	2,334	16.56%
13	All other non-current assets	2,473	1,880	593	31.54%
14	Total non-current assets	99,859	94,558	5,301	5.61%
15	Total assets	131,901	123,992	7,909	6.38%
16					
17	Short-term borrowings	12,118	9,629	2,489	25.85%
18	Trade and other payables	18,629	17,038	1,591	9.34%
19	Accruals and deferred income	3,855	3,673	182	4.96%
20	Provisions	620	564	56	9.93%
21	All other current liabilities	2,295	2,417	(122)	(5.05%)
22	Total current liabilities	37,517	33,321	4,196	12.59%
23	Long-term borrowings	11,091	11,601	(510)	(4.40%)
24	Provisions and other payables	5,027	4,330	697	16.10%
25	All other non-current liabilities	12,285	10,754	1,531	14.24%
26	Total non-current liabilities	28,403	26,685	1,718	6.44%
27	Total liabilities	65,920	60,006	5,914	9.86%
28	Share capital	311	319	(8)	(2.51%)
29	Treasury shares	(990)	(7,489)	(6,499)	(86.78%)
30	Retained earnings	82,870	88,014	(5,144)	(5.84%)
31	Other equity items	(16,210)	(16,858)	(648)	(3.84%)
32	Total equity	65,981	63,986	1,995	3.12%
33	Total liabilities and equity	131,901	123,992	7,909	6.38%
34					

From total assets perspective, you can see that Nestlé has grown 6.38%, from CHF 123,992 to CHF 131,901, as a result of increases in both current assets and non-current assets. In particular, cash has the biggest percentage increase of 63.60%. Short-term borrowings increased by 25.85%, accompanied by a decrease in long-term borrowings of 4.40%. Note that the largest decrease in quantum is retained earnings (CHF 5,144), but percentage wise, it has only dropped by 5.84%. This is why you will need to balance your review of horizontal analysis between the quantum and percentage since a large base may result in a smaller percentage change, and similarly, a small base may result in a very big percentage change.

A word of caution: you should not show a percentage change when the numbers swing from negative to positive or vice versa. In such cases, while you can mathematically calculate the percentage difference, they are not meaningful and not shown. For example, if you had a loss of \$100 last year and you now have a profit of \$150, you can't possibly use a percentage to describe the change. For these instances, you should put the notation "n.m." to stand for not meaningful.

» Stop & Think

Have another look at Exhibits 12-2 and 12-3. Are the biggest percentage items always the most important items that cause changes in the Income Statement or Balance Sheet from one year to another? Can you always ignore small changes, such as anything that is less than 5%?

Answer:

The largest percentage change in Nestlé's Income Statement was 122.06% for "other trading and operating income" (CHF 249). However, this amount is only the fifth highest change by amount. In comparison, the third largest amount change was CHF 684 for increase in sales, but it was only 0.77% in percentage terms. The 0.77% change would probably be more significant for Nestlé's performance than the 122.06% change.

Similarly, on the Balance Sheet, the smallest percentage change was CHF 159 (1.30%) change in "trade and other receivables." However, this is likely going to be more important than say, a CHF 56 (9.93%) change in provisions.

Thus, smaller percentage changes do not always mean they can be ignored. Similarly, a percentage change may be very substantial just because there was a small base.

Trend Analysis

Trend percentages are a form of horizontal analysis. Trends indicate the direction a business is taking. How have revenues changed over a five-year period? What trend does net income show? These questions can be answered by trend percentages over a representative period, such as the most recent five years.

Trend percentages are computed by selecting a base year whose amounts are set equal to 100%. The amount for each following year is stated as a percentage of the base amount. To compute a trend percentage, divide an item for a later year by the base-year amount.

$$\text{Trend \%} = \frac{\text{Any year's amount}}{\text{Base year's amount}}$$

Remember that income from operations (or operating profit) is often viewed as the primary measure of a company's earnings quality. This is because operating income represents a company's best predictor of the future net inflows from its core business units. Net income from operations is often used in estimating the current value of the business.

Nestlé's operating profit for the last 5 years is as follows:

(In millions, CHF)	2016	2015	2014	2013	2012
Operating profit	13,163	12,408	10,905	12,408	13,388

We want to calculate a trend for the five-year period 2012 through 2016. The first year in the series (2012) is set as the base year. Trend percentages are computed by dividing each year's amount by the 2012 amount. The resulting trend percentages follow (2012 = 100%):

	2016	2015	2014	2013	2012
Operating profit	98	93	81	98	100

Overall, Nestlé's operating income has been lower than 2012 (the highest in the five-year series). It dropped to a low of 81% in 2014 but has steadily climbed back to the same level as 2013.

You can perform a trend analysis on any item you consider important. Trend analysis using Income Statement data is widely used for predicting the future. Horizontal and trend analyses highlight changes over time. It is a basic technique that will get you started in financial statement analysis.

Vertical Analysis

Vertical analysis (or component analysis) shows the relationship of financial-statement items relative to a total, which is the 100% figure. All items on the particular financial statement are reported as a percentage of the base. For the Income Statement, total revenue (sales) is usually the base. Suppose under normal conditions a company's net income is 8% of revenue. A drop to 6% may cause the company's share price to fall. Let's apply this to Nestlé's Income Statement.

Exhibit 12-4 shows the vertical analysis of Nestlé's Income Statement as a percentage of revenue.

Exhibit 12-4 Vertical Analysis of Nestlé's Income Statements

A1					
	A	B	C	D	E
	Nestlé Group				
	Consolidated Income Statement (Adapted)				
	For the year ended Dec. 31				
1					
2	(In millions of Swiss Francs, CHF)	2016	% sales	2015	% sales
3	Sales	89,469	100.00%	88,785	100.00%
4	Other revenue	317	0.35%	298	0.34%
5	Cost of goods sold	(44,199)	(49.40%)	(44,730)	(50.38%)
6	Distribution expenses	(8,059)	(9.01%)	(7,899)	(8.90%)
7	Marketing and administration expenses	(21,485)	(24.01%)	(20,744)	(23.36%)
8	Research and development costs	(1,736)	(1.94%)	(1,678)	(1.89%)
9	Other trading and operating income	453	0.51%	204	0.23%
10	Other trading and operating expenses	(1,597)	(1.78%)	(1,828)	(2.06%)
11	Operating profit	13,163	14.71%	12,408	13.98%
12	Financial income	121	0.14%	101	0.11%
13	Financial expense	(758)	(0.85%)	(725)	(0.82%)
14	Taxes	(4,413)	(4.93%)	(3,305)	(3.72%)
15	Income from associates and joint ventures	770	0.86%	988	1.11%
16	Profit for the year	8,883	9.93%	9,467	10.66%
17					

Some observations from the above vertical analysis are as follows:

- Cost of sales was 50.38% of revenue in 2015, but it has decreased a percentage point to 49.40% in 2016. This indicates Nestlé's ability to manage cost of sales.
- In general, distribution expenses, marketing and administration expenses, and research and development costs were slightly higher in 2016 compared to 2015 as a percentage of sales.
- Consequently, the operating profit has increased from 14% to 15%. One percentage point may look small, but it translates to a 700 million CHF increase in operating profit for Nestlé.
- As a percentage of sales, taxes have gone up from 3.72% to 4.93%.
- Overall, for each CHF of sales in 2016, Nestlé made about 10% in net profit, and this was slightly down from 2015.

We could do a similar vertical analysis on the Balance Sheet. If you want to see the composition of total assets, then let total assets be 100% and express all assets as a percentage of total assets. However, if you are zooming in on the composition of current assets, then let current assets be 100% and examine only components of current assets. A common way to visualize

» Stop & Think

Perform a vertical analysis for the common-size percentages for the following Income Statement:

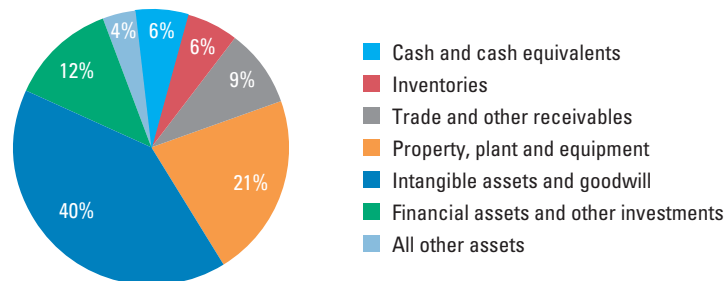
Net sales.....	\$150,000
Cost of goods sold.....	60,000
Gross profit.....	90,000
Operating expense.....	40,000
Operating income.....	50,000
Income tax expense.....	15,000
Net income.....	\$ 35,000

Answer:

Net sales.....	100%	(= \$150,000 ÷ \$150,000)
Cost of goods sold.....	40	(= \$ 60,000 ÷ \$150,000)
Gross profit.....	60	(= \$ 90,000 ÷ \$150,000)
Operating expense.....	27	(= \$ 40,000 ÷ \$150,000)
Operating income.....	33	(= \$ 50,000 ÷ \$150,000)
Income tax expense.....	10	(= \$ 15,000 ÷ \$150,000)
Net income.....	23%	(= \$ 35,000 ÷ \$150,000)

vertical analysis is to depict the items as a pie chart. For a pie chart to be effective, you would want to show no more than 7 slices and combine all other smaller items together. Exhibit 12-5 shows an example of component analysis of Nestlé's assets using a pie chart.

Exhibit 12-5 Vertical Analysis of Nestlé's Assets Using a Pie Chart



PREPARE COMMON-SIZE FINANCIAL STATEMENTS

Benchmarking

As you have seen throughout the textbook, when we do financial analysis, we normally benchmark the financial numbers and ratios to other similar companies. **Benchmarking** simply means comparing one entity to another. Usually benchmarks are selected because they direct competitors in the same industry or market, peers in the broader market, or just any other “aspiration” entities. You may be running a small retail shop, but if you aspire to do just as well as Amazon or department stores like Marks and Spencer, then go ahead, you can always benchmark your own company's

2 Perform common-size financial statements

performance against specific companies. The most important aspect of benchmarking, however, is that it gives you context in which you could interpret your data.



A Closer Look

You could also research industry averages on various websites. For example, <http://www.bizstats.com/industry-financials.php> offers some basic financial ratio information from 251 industries and sub-sectors. In addition, your university library may have subscriptions to Dun & Bradstreet's *Industry Norms and Key Business Ratios*, Standard & Poor's *Capital IQ*, or Risk Management Association's *Annual Statement Studies*. There are also many others, but they typically require you to subscribe or purchase the information. In this chapter, relevant industry benchmarks are obtained from S&P's *Capital IQ* at the time of writing.

Suppose you are a financial analyst for a large investment bank. You are considering investing in one of two different companies in the same industry, say Nestlé, Unilever, or P&G. A direct numerical comparison of these companies' financial statements is not meaningful, in part because of differing currencies and size differences. One company may be so much larger than the others. One of the techniques you could use, called **common-size financial statements**, is basically an extension of vertical analysis.

We can modify Exhibit 12-4 to report only percentages (no currency amounts) for two or more companies. When compared side-by-side, such common-size financial statements aid the comparison of different companies because all amounts are stated in percentages, thus expressing the financial results of each comparative company in terms of a common denominator. Currency and size differences are eliminated when you do a common-size comparison.

Exhibit 12-6 presents the common-size Income Statements of Nestlé, Unilever, and P&G. To make common-size comparisons meaningful, you often have to synchronize the items in the two (or more) financial statements and simplify. For example, one company may separate selling and distribution expenses from administrative expenses, but another may combine them as selling, general, and administrative expenses. Another difference could be where one company provides analysis of its expenses by function and another company provides it by nature (we discussed function vs. nature in Chapter 4). Sometimes, you may only be able to compare the key line items. Exhibit 12-6 offers the broad expense categories for our comparisons of the three FMCG companies.

Note that Nestlé reports in Swiss Francs (CHF), Unilever in euros (EUR) and P&G in US dollars (USD), and yet we are able to arrive at some comparative numbers using common-size analysis. This is one of the major advantages of using percentages rather than comparing actual amounts.

Exhibit 12-6 Common-size Income Statements

A1				
	A	B	C	D
1	Common-size Income Statements	Nestlé 2016	Unilever 2016	P&G 2016
2	Sales	100.00%	100.00%	100.00%
3	Cost of goods sold	(49.40%)	(51.19%)	(50.40%)
4	Gross profit	50.60%	48.81%	49.60%
5	Other operating income and expenses	(35.89%)	(34.01%)	(29.02%)
6	Operating profit	14.71%	14.80%	20.58%
7	Interest expense	(0.85%)	(1.11%)	(0.89%)
8	Tax expense	(4.93%)	(3.65%)	(5.12%)
9	Other non-operating income and expenses	1.00%	0.48%	1.66%
10	Profit for the year	9.93%	10.52%	16.24%

As you can see from Exhibit 12-6, Nestlé has the highest gross profit margin, but the lowest operating profit margin and net profit margin. This means both its operating and nonoperating income and expenses are higher than Unilever and P&G. Looking at the bottom-line performance, we can see that Unilever is slightly better than Nestlé, but P&G's performance is about 5 percentage points better than Nestlé and Unilever.

Mid-Chapter Summary Problem

Perform a horizontal analysis and a vertical analysis of the comparative Income Statement of Hard Rock Products, Inc., which makes metal detectors. State whether 20X7 was a good year or a bad year, and give your reasons.

A1			
	A	B	C
	Hard Rock Products, Inc.		
	Comparative Income Statements		
	Years Ended December 31, 20X7 and 20X6		
1			
2		20X7	20X6
3	Total revenues	\$ 275,000	\$ 225,000
4	Expenses:		
5	Cost of goods sold	194,000	165,000
6	Engineering, selling, and administrative expenses	54,000	48,000
7	Interest expense	5,000	5,000
8	Income tax expense	9,000	3,000
9	Other expense (income)	1,000	(1,000)
10	Total expenses	263,000	220,000
11	Net income	\$ 12,000	\$ 5,000
12			

Answer

The horizontal analysis shows that total revenues increased 22.2%. This was greater than the 19.5% increase in total expenses, resulting in a 140% increase in net income.

A1					
	A	B	C	D	E
	Hard Rock Products, Inc.				
	Horizontal Analysis of Comparative Income Statements				
	Years Ended December 31, 20X7 and 20X6				
1					
2		20X7	20X6	Increase (Decrease)	
3	Total revenues	\$ 275,000	\$ 225,000	\$ 50,000	22.2%
4	Expenses:				
5	Cost of goods sold	194,000	165,000	29,000	17.6
6	Engineering, selling, and				
7	administrative expenses	54,000	48,000	6,000	12.5
8	Interest expense	5,000	5,000	—	—
9	Income tax expense	9,000	3,000	6,000	200.0
10	Other expense (income)	1,000	(1,000)	2,000	n.m.*
11	Total expenses	263,000	220,000	43,000	19.5
12	Net income	\$ 12,000	\$ 5,000	\$ 7,000	140.0%
13					

*n.m., not meaningful. Percentage changes are typically not computed for shifts from a negative to a positive amount and vice versa.

The vertical analysis shows decreases in the percentages of net sales consumed by the cost of goods sold (from 73.3% to 70.5%) and by the engineering, selling, and administrative expenses (from 21.3% to 19.6%). Because these two items are Hard Rock's largest dollar expenses, their percentage decreases are quite important. The relative reduction in expenses raised 20X7 net income to 4.4% of sales, compared with 2.2% the preceding year. The overall analysis indicates that 20X7 was significantly better than 20X6.

A1					
A					
B					
C					
D					
E					
Hard Rock Products, Inc.					
Vertical Analysis of Comparative Income Statements					
Years Ended December 31, 20X7 and 20X6					
20X7					
20X6					
Amount					
Percent					
Amount					
Percent					
1	Total revenues	\$ 275,000	100.0%	\$ 225,000	100.0%
2	Expenses:				
3	Cost of goods sold	194,000	70.5	165,000	73.3
4	Engineering, selling, and administrative expenses	54,000	19.6	48,000	21.3
5	Interest expense	5,000	1.8	5,000	2.2
6	Income tax expense	9,000	3.3	3,000	1.4**
7	Other expense (income)	1,000	0.4	(1,000)	(0.4)
8	Total expenses	263,000	95.6	220,000	97.8
9	Net income	\$ 12,000	4.4%	\$ 5,000	2.2%
10					
11					
12					

**Number rounded up.

PERFORM FINANCIAL RATIO ANALYSIS TO MAKE BUSINESS DECISIONS

3 Perform financial ratio analysis to make business decisions



Financial ratios are a major tool of financial analysis. We have discussed the use of many ratios in financial analysis in various chapters throughout the book. A ratio expresses the relationship of one number to another. Suppose your Balance Sheet shows current assets of \$100,000 and current liabilities of \$50,000. The ratio of current assets to current liabilities is \$100,000 to \$50,000. We can express this ratio as 2 to 1, or 2:1, or some may even say current ratio is at 200%. In any case, what we really mean is that the current ratio is 2.

Many companies include selected financial ratios in a special section of their annual reports or post them online on their "investor relation" websites. For example, Nestlé's corporate website, under the Investor tab, provides a link to its "Annual Review" where a number of ratios are shown and discussed. Similarly, on the Unilever website (<https://www.unilever.com/investor-relations/annual-report-and-accounts/>) you can see a list of key ratios highlighted. Alternatively, you may visit various information websites, such as Moody's, Standard & Poor's, Risk Management Association, Reuters, and others, for their coverage of companies' news and financial information, including financial ratios. You are more likely to find more coverage on larger multinational companies than on smaller ones. For example, you can check out Reuters' coverage of Nestlé at <http://www.reuters.com/finance/stocks/financialHighlights?symbol=NESN.S>.

The financial ratios we discuss in this chapter are classified as follows:

1. Efficiency ratios
2. Financial strength ratios
3. Profitability ratios
4. Investment ratios

Sometimes you may find different ratio classifications and even slightly different formulas for the ratios we are about to discuss. For example, Reuters classify ratios into efficiency, effectiveness, financial strength, profitability, and valuation ratios. Don't be alarmed by any differences; as long as you calculate the ratios consistently, there is always value to financial

statement analysis. A weighing scale that is slightly inaccurate can still tell you which item in a room is the heaviest or lightest.

Exhibit 12-7 gives the Income Statement and Balance Sheet data of Ikon Furniture, an independent furniture and homewares retailer. We will work with this exhibit as our running example on how to perform financial statement analysis and compare the financial ratios to an industry average. At the end of each section, we will also show you the financial ratios of Nestlé for discussion.

Exhibit 12-7 | Comparative Financial Statements

A1			
	A	B	C
1	Ikon Furniture, Inc.		
2	Comparative Income Statements		
3	Years Ended December 31, 20X7 and 20X6		
4		20X7	20X6
5	Net sales	\$ 858,000	\$ 803,000
6	Cost of goods sold	513,000	509,000
7	Gross profit	345,000	294,000
8	Operating expenses:		
9	Selling expenses	126,000	114,000
10	General expenses	118,000	123,000
11	Total operating expenses	244,000	237,000
12	Income from operations	101,000	57,000
13	Interest revenue	4,000	0
14	Interest (expense)	(24,000)	(14,000)
15	Income before income taxes	81,000	43,000
16	Income tax expense	33,000	17,000
17	Net income	\$ 48,000	\$ 26,000

A1			
	A	B	C
1	Ikon Furniture, Inc.		
2	Comparative Balance Sheets		
3	December 31, 20X7 and 20X6		
4		20X7	20X6
5	Assets		
6	Current Assets:		
7	Cash	\$ 29,000	\$ 32,000
8	Accounts receivable, net	114,000	85,000
9	Inventories	113,000	111,000
10	Prepaid expenses	6,000	8,000
11	Total current assets	262,000	236,000
12	Long-term investments	18,000	9,000
13	Property, plant and equipment, net	507,000	399,000
14	Total assets	\$ 787,000	\$ 644,000
15	Liabilities		
16	Current Liabilities:		
17	Notes payable	\$ 42,000	\$ 27,000
18	Accounts payable	73,000	68,000
19	Accrued liabilities	27,000	31,000
20	Total current liabilities	142,000	126,000
21	Long-term debt	289,000	198,000
22	Total liabilities	431,000	324,000
23	Shareholders' Equity		
24	Share capital, no par	186,000	186,000
25	Retained earnings	170,000	134,000
26	Total shareholders' equity	356,000	320,000
27	Total liabilities and shareholders' equity	\$ 787,000	\$ 644,000

Efficiency Ratios

For companies that buy or make goods for sale, the ability to sell inventory and collect receivables is critical. In this section, we discuss a number of ratios that measure an entity’s ability to collect cash. We have previously introduced some of the ratios here in earlier chapters (receivable turnover, inventory turnover, and payable turnover in Chapters 5, 6, and 9). Let’s start with the components of a cash conversion cycle.

Inventory Turnover. Companies generally strive to sell their inventory as quickly as possible. The faster inventory sells, the sooner cash comes in. **Inventory turnover** (discussed in Chapter 6) measures the number of times a company sells its average level of inventory during a year. A fast turnover indicates ease in selling inventory; a low turnover indicates difficulty. A value of 6 means that the company’s average level of inventory has been sold six times during the year, and that’s usually better than a turnover of 3. But too high a value can mean that the business is not keeping enough inventory on hand, which can lead to lost sales if the company can’t fill orders. Therefore, a business strives for the most profitable rate of turnover, not necessarily the *highest* rate.

To compute inventory turnover, divide the cost of goods sold by the average inventory for the period. We use the cost of goods sold—not sales—in the computation because both cost of goods sold and inventory are stated at cost. Ikon Furniture’s inventory turnover for 20X7 is:

Formula	<i>Ikon’s Inventory Turnover</i>	Industry Average
Inventory turnover = $\frac{\text{Cost of goods sold}}{\text{Average inventory}}$	$\frac{\$513,000}{\$112,000} = 4.6$	4.2

Cost of goods sold comes from the Income Statement (Exhibit 12-7). Average inventory is the average of beginning (\$111,000) and ending inventory (\$113,000) (see the Balance Sheet, Exhibit 12-7). Inventory turnover varies widely with the nature of the business. For example, a fast-food chain restaurant, such as Pizza Hut, Taco Bell, KFC, or Long John Silver’s, is likely to have a high turnover ratio because it turns around its inventory into sales in a short time. This may be a necessity because food spoils quickly. On the other hand, a furniture retailer turns its inventory over only a few times a year, as most of the inventory is kept for a while on shop floors and warehouses, waiting for customers to purchase. In this case, Ikon’s inventory turnover is slightly above the industry average. To evaluate inventory turnover, compare the ratio over time. A sharp decline suggests the need for corrective action.

Inventory turnover can also be expressed in number of days. If you divide 365 days by Ikon’s inventory turnover ratio of 4.6, this is equal to about 79 days, compared to the industry average of 87 days. This ratio is called the **inventory resident period** (or days supply on hand, days inventory on hand, or something to that effect).

Accounts Receivable Turnover. Receivable turnover measures the ability to collect cash from customers. In general, the higher the ratio, the better. A low receivable turnover indicates ineffectiveness in collecting dues from customers. However, a receivable turnover that is too high may indicate that credit is too tight, and that may cause you to lose sales to good customers.

To compute accounts receivable turnover, divide net sales by average net accounts receivable. The ratio tells how many times during the year average receivables were turned into cash. Note that we would normally exclude non-trading revenue and receivables. For example, we would only include revenue from trading activities and not interest revenue or dividend income. Similarly, we would only include trade receivables and not other receivables such as loans to employees. Ikon Furniture’s accounts receivable turnover ratio for 20X7 is:

Formula	<i>Ikon’s Accounts Receivable Turnover</i>	Industry Average
Accounts receivable turnover = $\frac{\text{Sales Revenue}}{\text{Average accounts receivable}}$	$\frac{\$858,000}{\$99,500} = 8.6$	6.2

Average net accounts receivable can be derived by adding beginning (\$85,000) and ending receivables (\$114,000), then dividing by 2. Ikon's receivable turnover of 8.6 times per year is a little faster than the industry average. Why the faster collection? Ikon may be a hometown store that sells to local people and businesses who typically settle their bills on time.

Similarly, we can also convert receivable turnover into days and refer to it as the **receivable collection period**, also known as days sales outstanding, or days sales in receivables, or something similar. For Ikon, on average, it takes 42 days to collect cash from its receivables. This is better than the home furnishing industry's average of 59 days.

Accounts Payable Turnover. Businesses buy their supplies and raw materials on credit, and take time to pay their accounts payable. A high account payable turnover ratio means a business pays its suppliers very quickly, and a low payable turnover means a longer time period for payments to suppliers. Generally, a lower payable turnover is better than a higher one, as the business is making full use of the credit terms extended by its creditors. However, a business can't stretch the payable period too far because no one would supply the business if it continued to be delinquent on its payments.

To compute **payable turnover**, divide cost of goods sold by the average accounts payable for the period. Again, we would only include trade-related payables and exclude items such as interest payable, short-term loan, tax payable, and so forth. Average accounts payable for the period is \$73,000 + \$60,000 divided by 2. Ikon Furniture's payable turnover for 20X7 is:

Formula	<i>Ikon's Accounts Payable Turnover</i>	Industry Average
Accounts payable turnover = $\frac{\text{COGS}}{\text{Average net accounts payable}}$	$\frac{\$513,000}{\$70,500} = 7.3$	3.7

It seems that Ikon pays its creditors faster than the industry average. Perhaps as a small store, it was not given as liberal a credit term as other major players in the home furnishing industry. If we express this ratio in days, the **payable outstanding period** (or days payable outstanding) is 50 days ($365/7.3$), whereas the industry average is about 99 days ($365/3.7$).

Cash Conversion Cycle. If we put the inventory resident period, receivable collection period, and payable collection period together, we can get a rough idea of how long it takes for a business to sell its inventory, collect payments, and make its payments to suppliers. This is what we call the **cash conversion cycle**.

Ikon's cash conversion cycle is thus inventory resident period + receivable collection period – payable outstanding period, or $79 + 42 - 50 = 71$ days. The home furnishings' cash conversion cycle is much faster at $87 + 59 - 99 = 47$ days, primarily due to the long payable outstanding period. In general, a shorter cycle is better than a longer cycle; in fact it can even be a negative number. Here's an example of HP (Hewlett-Packard)'s disclosure on its cash conversion cycle. Note that HP uses slightly different terminology and 90-day averages instead of 2-year averages that we use in this chapter.

ADAPTED EXCERPTS FROM HP INC'S ANNUAL FILING

Management utilizes current cash conversion cycle information to manage our working capital level. The table below presents the cash conversion cycle as of October 31, 2016, and October 31, 2015.

	As of October 31		
	2016	2015	2014
Days of sales outstanding in accounts receivable	30	35	33
Days of supply in inventory	39	39	35
Days of purchases outstanding in accounts payable	(98)	(93)	(86)
Cash conversion cycle	(29)	(19)	(18)

continued on the following page

The sum of days' sales outstanding in accounts receivable ("DSO") and days' supply in inventory ("DOS") less days' purchases outstanding in accounts payable ("DPO") is known as Cash Conversion Cycle.

The DSO is used to account for the average number of days of outstanding receivables. We can calculate the DSO by dividing ending accounts receivable, net of allowance for doubtful accounts, by a 90-day net revenue average. The DOS also accounts for the average number of days from procurement to the sale of a product. It is computed by dividing the ending inventory by a 90-day average of cost of goods sold.

The DPO is used to compute the average number of days for outstanding accounts payable balances. It is calculated by dividing ending accounts payable by a 90-day average of cost of goods sold.

Source: HP Inc., Annual Filing 2016, page 45–46

» Stop & Think

Use the following information to calculate Nestlé's Cash Conversion Cycle for 2016 and 2015. What is your assessment of Nestlé's trading efficiency?

Nestlé (in CHF Millions)	2016	2015	2014
Sales	89,469	88,785	91,612
COGS	44,199	44,730	47,553
Inventories	8,401	8,153	9,172
Trade receivables	10,023	9,696	10,283
Trade payables	18,629	17,038	17,437

Answer:

Inventory turnover and resident period 2016 = $44,199 / \text{average of } 8,401 \text{ and } 8,153 = 5.34 \text{ times}$; and $365 / 5.34 = 68.35 \text{ days}$

Inventory turnover and resident period 2015 = $44,730 / \text{average of } 8,153 \text{ and } 9,172 = 5.16 \text{ times}$; and $365 / 5.16 = 70.69 \text{ days}$

Receivables turnover and collection period 2016 = $89,469 / \text{average of } 10,023 \text{ and } 9,696 = 9.07 \text{ times}$; and $365 / 9.07 = 40.22 \text{ days}$

Receivables turnover and collection period 2015 = $88,785 / \text{average of } 9,696 \text{ and } 10,283 = 8.89 \text{ times}$; and $365 / 8.89 = 41.07 \text{ days}$

Payable turnover and payment period 2016 = $44,199 / \text{average of } 18,629 \text{ and } 17,038 = 2.48 \text{ times}$; and $365 / 2.48 \text{ times} = 147.27 \text{ days}$

Payable turnover and payment period 2015 = $44,730 / \text{average of } 17,038 \text{ and } 17,437 = 2.59 \text{ times}$; and $365 / 2.59 \text{ times} = 140.66 \text{ days}$

Cash conversion cycle 2016 = $68.35 \text{ days} + 40.22 \text{ days} - 147.27 \text{ days} = -38.70 \text{ days}$

Cash conversion cycle 2015 = $70.69 \text{ days} + 41.07 \text{ days} - 140.66 \text{ days} = -28.91 \text{ days}$.

Nestlé has a very good cash conversion cycle. The time it took to pay suppliers is longer than the time it took to sell inventory and collect cash, resulting in a negative CCC. There is a limit, however, on how long Nestlé can practically delay payment to its suppliers. Suppliers may increase their prices or even refuse to supply Nestlé if they have to wait for a long time for payments.

Asset Turnover Ratio. Another way to examine efficiency would be to assess the amount of resources used to generate sales or revenue. This can be done on a total assets basis, or sometimes on a fixed assets (i.e., PPE) basis. This ratio is calculated by dividing sales or revenue by average total assets.

Formula	<i>Ikon's Asset Turnover</i>		Industry Average
	20X7	20X6	
Asset Turnover = $\frac{\text{Sales}}{\text{Average total assets}}$	$\frac{\$858,000}{\$715,500} = 1.2$	$\frac{\$803,000}{\$644,000} = 1.2$	0.9

*Using 20X6 ending figures only

Ikon has been able to generate \$1.2 worth of sales for every \$1 of assets in 20X6 and 20X7. Its assets turnover is higher than the industry average of 0.9.

Financial Strength Ratios

Financial strength ratios are indicators of an entity's abilities to meet its financial obligations, either in the short-term or the long-term. Short-term indicators are usually called **liquidity ratios** and long-term ones are usually called **solvency ratios**. Let's look at these indicators in turn.

In any financial year, a business has current assets at its disposal and has current liabilities to settle. **Working capital** is the term we use to describe what a business has to work with during the year, i.e., current assets less current liabilities. Generally, the higher the working capital, the better. Too much of it, however, means the business is not utilizing its current assets effectively, as cash, inventory, and receivables do not earn much income. Some businesses may operate on very little working capital, especially if they have a very strong cash conversion cycle. Consider two companies with equal working capital:

	<i>Company</i>	
	Jones	Smith
Current assets.....	\$100,000	\$200,000
Current liabilities	<u>50,000</u>	<u>150,000</u>
Working capital	<u>\$ 50,000</u>	<u>\$ 50,000</u>

Both companies have working capital of \$50,000, but Jones' working capital is as large as its current liabilities. Smith's working capital is only one-third as large as current liabilities. Jones is in a better position because its working capital is a higher percentage of current liabilities. Two decision-making tools based on current assets and current liabilities data are the *current ratio* and the *quick ratio*.

Current Ratio. The most common ratio evaluating current assets and current liabilities is the **current ratio**, which is simply current assets divided by current liabilities. The current ratio measures the ability to pay current liabilities with current assets (as discussed in Chapter 5).

The current ratios of Ikon Furniture, Inc., at December 31, 20X7 and 20X6, follow, along with the average for the home furnishing industry:

Formula	<i>Ikon's Current Ratio</i>		Industry Average
	20X7	20X6	
Current ratio = $\frac{\text{Current assets}}{\text{Current liabilities}}$	$\frac{\$262,000}{\$142,000} = 1.85$	$\frac{\$236,000}{\$126,000} = 1.87$	1.6

Ikon's current ratio decreased slightly during 20X7. In general, a higher current ratio indicates a stronger financial position. The business has sufficient current assets to maintain its operations. What is an acceptable current ratio? The answer depends on the industry. The

norm for companies in most industries is around 1.5, as reported by the Risk Management Association. Ikon Furniture's current ratio of 1.85 is better than the home furnishing industry average of 1.6.

Quick Ratio. A more refined version of current ratio is known as the **quick ratio** (also known as the **acid-test ratio**). It tells us whether a business could pass the "acid test" of paying all its current liabilities if they came due immediately. The quick ratio uses a narrower base to measure liquidity than the current ratio does.

To compute the quick ratio, we add cash, short-term investments, and net current receivables (accounts and notes receivable, net of allowances) and divide by current liabilities. Inventory is excluded because they are less liquid, as businesses may be unable to convert inventory to cash immediately. This is why sometimes you see this shortcut in measuring quick ratio: $(\text{Current Assets} - \text{Inventory}) \div \text{Current Liabilities}$.

Ikon Furniture's quick ratios for 20X7 and 20X6 follow:

Formula	<i>Ikon's Quick Ratio</i>		Industry Average
	20X7	20X6	
Quick ratio = $\frac{\text{Cash + Short-term investments} + \text{Net current receivables}}{\text{Current liabilities}}$	$\frac{\$29,000 + \$0 + \$114,000}{\$142,000} = 1.01$	$\frac{\$32,000 + \$0 + \$85,000}{\$126,000} = 0.93$	1.3

The company's quick ratio improved during 20X7 and is lower than the industry average. It seems like Ikon is holding more inventory than other companies in the industry.

» Stop & Think

Use the following information to calculate Nestlé's short-term financial strength ratios for 2016 and 2015. What is your assessment of Nestlé's liquidity?

Nestlé (in CHF Millions)	2016	2015
Current assets	32,042	29,434
Current liabilities	37,517	33,321
Inventories	8,401	8,153

Answer:

Working capital 2016 = $32,042 - 37,517$ = negative working capital of CHF 5,475 million.

Working capital 2015 = $29,434 - 33,321$ = negative working capital of CHF 3,887 million.

Current ratio 2016 = $32,042 / 37,517 = 0.85$

Current ratio 2015 = $29,434 / 33,321 = 0.88$

Quick ratio 2016 = $(32,042 - 8,401) / 37,517 = 0.63$

Quick ratio 2015 = $(29,434 - 8,153) / 33,321 = 0.64$

Nestlé is working under a negative capital and has a current ratio well below 1.5. This is usually an indicator of financial weakness. However, recall that Nestlé has a negative cash conversion cycle, which makes it less necessary to have large current assets to meet its current liabilities. We could also calculate quick ratio using its alternative form if we had the breakdown of the items instead of just the total current assets: $\text{Cash} + \text{Short-term Investment} + \text{Receivables} / \text{Current Liabilities}$.

Debt Ratio. Suppose you are a bank loan officer and you have received \$500,000 loan applications from two similar companies. The first company already owes \$600,000, and the second owes only \$250,000. Which company gets the loan? All else being equal, you will probably approve the loan application from Company 2, because it owes less.

This relationship between total liabilities and total assets is called the **debt ratio**, which gives an indication of the degree of **leverage** or **gearing** of a company. The debt ratio tells us the proportion of assets financed with debt (see Chapter 9). A debt ratio of 1 reveals that debt has financed all the assets. A debt ratio of 0.50 means that debt finances half the assets. The higher the debt ratio, the greater the pressure to pay interest and principal. The lower the ratio, the lower the risk. You can also express debt ratio as a percentage.

The debt ratios for Ikon Furniture in 20X7 and 20X6 follow:

Formula	<i>Ikon's Debt Ratio</i>		Industry Average
	20X7	20X6	
Debt ratio = $\frac{\text{Total liabilities}}{\text{Total assets}}$	$\frac{\$431,000}{\$787,000} = 0.55$	$\frac{\$324,000}{\$644,000} = 0.50$	0.62

Capital IQ reports that the average debt ratio for home furnishing companies is about 0.62 (or 62%), with relatively little variation from company to company. Ikon's 55% debt ratio indicates a fairly low-risk debt position compared with the industry average of 62%.

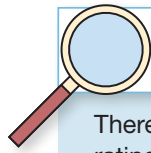
An alternative way to describe the level of an entity's leverage is the **debt-to-equity** ratio. Recall that the accounting equation is Assets = Liabilities + Equity. If you know the debt ratio (i.e., liabilities over assets), you can easily work out an entity's debt-to-equity ratio. If total assets are \$787,000 and total liabilities are \$431,000, equity must be \$356,000. The debt-to-equity ratio is thus \$431,000 over \$356,000, or 1.21 times. A debt-ratio of 0.55 is equivalent to a debt-to-equity ratio of 1.21.

Times-Interest-Earned Ratio. Analysts use another ratio—the **times-interest-earned ratio** (or interest coverage ratio)—to relate income to interest expense. To compute the times-interest-earned ratio, divide income from operations (operating income) by interest expense. This ratio measures the number of times operating income can cover interest expense and is also called the **interest-coverage ratio**. A high ratio indicates ease in paying interest; a low value suggests difficulty.

Ikon's times-interest-earned ratios are:

Formula	<i>Ikon's Times-Interest-Earned Ratio</i>		Industry Average
	20X7	20X6	
Times-interest-earned ratio = $\frac{\text{Income from operations}}{\text{Interest expense}}$	$\frac{\$101,000}{\$24,000} = 4.21$	$\frac{\$57,000}{\$14,000} = 4.07$	5.8

The company's times-interest-earned ratio increased in 20X7. This is a favorable sign but it is lower than the industry's average interest coverage ratio. This means there are some risks that any shortfall in profitability may negatively affect Ikon's ability to pay interest to its creditors.



A Closer Look

There are many other versions of these ratios, depending on which textbook, website, rating agency, or financial institution you read or follow. For example, some would consider “debt” in debt ratio to be more specific than total liabilities. They would prefer to interpret debt as “interest-bearing liabilities,” and exclude other items such as accounts payable, accrued liabilities, unearned revenue, and so forth because these liabilities do not bear interest. When you think of your own “debt,” you would naturally think of your tuition loan or motor vehicle loan, and not your outstanding telephone bill, or that your friend paid for your lunch last week!

Similarly, there are numerous additional refinements one could employ. For example, when interest payments are capitalized (under IAS 23—Borrowing Costs, briefly discussed in Chapter 7), these payments will not be shown as an expense. These analysts, in calculating times-interest-earned, prefer to add any capitalized interest to the interest expense shown on the Income Statement on the ratio’s denominator. To them, the ability to pay interest does not depend on whether the interest is expensed or capitalized.

» Stop & Think

Use the following information to calculate Nestlé’s long-term financial strength ratios for 2016 and 2015. What is your assessment of Nestlé’s solvency?

Nestlé (in CHF Millions)	2016	2015
Total assets	131,901	123,992
Total liabilities	65,920	60,006
Total equity	65,981	63,986
Operating profit	13,163	12,408
Interest expense	758	725

Answer:

Debt ratio 2016 and 2015 = $65,920/131,901 = 0.50$ times and $60,006/123,992 = 0.48$ times

Debt to equity ratio 2016 and 2015 = $65,920/65,981 = 1.00$ times and $60,006/63,986 = 0.94$ times

Times interest earned 2016 and 2015 = $13,163/758 = 17.37$ times and $12,408/725 = 17.11$ times

Long-term measures generally do not fluctuate much from one year to another unless there is a major change in an entity’s capital structure. Nestlé funds its assets from roughly equal amounts shareholder funds and creditors. Interest-bearing loans are a small part of total liabilities, which why it has a very high (and healthy) ability to pay interest expenses.

Profitability Ratios

The fundamental goal of a business is to earn a profit, and so the ratios that measure profitability are reported widely. Profitability ratios may be expressed in decimals or percentages. For example, 0.10 is equal to 10%, 0.25 equals 25%, etc. As investors are accustomed to seeing returns in percentages, we will use % for profitability ratios. You will also see that we use the term “return” for many of the profitability ratios. In this context, returns simply mean any measure of profit, usually net profit.

Gross Profit, Operating Profit, and Net Profit Margin. These ratios show the percentage of each sales dollar earned as gross, operating, and net profit. If the company does not explicitly have a line on its Income Statement as operating profit (or profit from operations), you can use earnings before interest and tax (EBIT) as a surrogate for operating profit. In fact, when you did the common-size analysis in Exhibit 12-6, you were already calculating operating profit margins and net profit margins.

The margin-related ratios for Ikon Furniture are:

Formula	<i>Ikon's margin-related ratios</i>		Industry Average
	20X7	20X6	
Gross Profit Margin = $\frac{\text{Gross profit}}{\text{Sales}}$	$\frac{\$345,000}{\$858,000} = 40.2\%$	$\frac{\$294,000}{\$803,000} = 36.6\%$	31.6%
Operating Profit Margin = $\frac{\text{Operating profit}}{\text{Sales}}$	$\frac{\$101,000}{\$858,000} = 11.8\%$	$\frac{\$57,000}{\$803,000} = 7.1\%$	7.9%
Net Profit Margin = $\frac{\text{Net profit}}{\text{Sales}}$	$\frac{\$48,000}{\$858,000} = 5.6\%$	$\frac{\$26,000}{\$803,000} = 3.2\%$	5.6%

Companies strive for a high rate of net income (sometimes referred to as **return on sales**). The higher the percentage, the more profit is being generated by sales dollars. Ikon Furniture's gross, operating, and net profit margins have improved in 20X7. Ikon was able to generate gross profit and operating profit margins above the industry averages, and its net profit margin matches the home furnishing industry's average.

Return on Total Assets. The **return on total assets** (ROA) (introduced in Chapter 7), measures a company's success in using assets to earn a profit. There are some variations in the numerator for this ratio. Some textbooks advocate the use of net profit, others use net profit but with interest expenses added back, or operating profit, or earnings before interest and tax. We will use net profit in this text. Average total assets is the denominator, which is the total of beginning and ending balances (\$787,000 and \$644,000) divided by 2. On occasions when we don't have access to beginning balances (e.g., we don't have Ikon's beginning balances for 20X6), we can use the ending balance instead of the average. We will show Ikon's ROA after discussing ROE in the next section.

Return on Equity. A popular measure of profitability is the **return on ordinary shareholders' equity**, often shortened to *return on equity* (ROE). This ratio shows the relationship between net income and ordinary shareholders' investment in the company—how much income is earned for every \$1 invested by the ordinary equity shareholders (see Chapter 10). Naturally, if a company does not have preference shares (like Ikon Furniture), we do not have to worry about excluding preference dividends and preference equity. We would simply take the total return figure and divide it by the average total equity in our calculations.



A Closer Look

You may be wondering why we specifically single out ordinary equity for this ratio. Recall that preference shares have a predetermined, fixed return in the form of preference dividends to the preference shareholders. Thus, after paying for the preference dividends, profits are available to ordinary shareholders.

For companies with preference shares, you will need to exclude preference dividends from the numerator and preference capital (both paid-in preference capital and any capital in excess of par for the preference capital). The formula to use when you are analyzing ROE for a company with preference shares will thus be:

$$\frac{\text{Net income} - \text{Preference dividend}}{\text{Average of total equity} - \text{Preference equity}}$$

Let's see Ikon Furniture's ROA and ROE for 20X7 and 20X6. For 20X6, we will use ending balances only as we do not have information about Ikon's 20X6 beginning balances (i.e., the ending 20X5 numbers) to calculate averages. Ikon Furniture has no preference shares and thus has no preference dividends. All of its equity is therefore ordinary. Average equity for 20X7 uses the beginning and ending balances $[(\$320,000 + \$356,000)/2 = \$338,000]$.

Formula	<i>Ikon's ROA and ROE</i>		Industry Average
	20X7	20X6	
Return on asset = $\frac{\text{Net profit}}{\text{Average total assets}}$	$\frac{\$48,000}{\$715,500} = 6.7\%$	$\frac{\$26,000}{\$644,000} = 4.0\%$	4.0%
Return on equity = $\frac{\text{Net profit}}{\text{Average equity}}$	$\frac{\$48,000}{\$338,000} = 14.2\%$	$\frac{\$26,000}{\$320,000} = 8.1\%$	8.6%

*Using 20X6 ending figures only

Ikon improved its ROA and ROE significantly in 20X7, generating 6.7% return on assets and 14.2% return on equity. The home furnishings' industry averages are only 4% and 8.6%, respectively. Observe that Ikon's return on equity (14.2%) is higher than its return on assets (6.7%). All businesses that have liabilities will have higher ROE than ROA. In fact, the higher the debt ratio, the higher the leverage. Companies that finance operations with significant debt are said to be highly leveraged. All is well when profits are greater than are necessary to meet interest payments. However, as the global financial crisis has taught us, highly leveraged companies are the first ones to fall when there is a drop in profitability. If revenues drop, debts must still be paid. Therefore, leverage is a double-edged sword. It increases profits during good times but compounds losses during bad times.

» Stop & Think

Use the following information to calculate Nestlé's profitability ratios for 2016 and 2015. What is your assessment of Nestlé's profitability?

Nestlé (in CHF Millions)	2016	2015	2014
Sales	89,469	88,785	91,612
Gross profit	45,270	44,055	44,059
Operating profit	13,163	12,408	10,905
Net profit	8,883	9,467	14,904
Total assets	131,901	123,992	133,450
Total equity	65,981	63,986	71,884

Answer:

Gross profit margin 2016 and 2015 = $45,270/89,469 = 50.60\%$ and $44,055/88,785 = 49.62\%$

Operating profit margin 2016 and 2015 = $13,163/89,469 = 14.71\%$ and $12,408/88,785 = 13.98\%$

Net profit margin 2016 and 2015 = $8,883/89,469 = 9.93\%$ and $9,467/88,785 = 10.66\%$

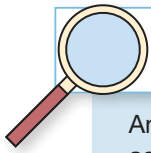
Return on assets 2016 = $8,883/\text{average of } 131,901 \text{ and } 123,992 = 6.94\%$

Return on assets 2015 = $9,467/\text{average of } 123,992 \text{ and } 133,450 = 7.35\%$

Return on equity 2016 = $8,883/\text{average of } 65,981 \text{ and } 63,986 = 13.67\%$

Return on equity 2015 = $9,467/\text{average of } 63,986 \text{ and } 71,884 = 13.94\%$

While Nestlé has done better in gross profit margin and operating profit margin, the lower net profit in 2016 has resulted in lower net profit margin, lower return on assets, and lower return on equity.



A Closer Look

Analysts use a technique called *DuPont Analysis* to better understand how ROE is achieved. This technique is made popular by the DuPont Corporation, which started using this analysis back in the 1920s. The DuPont Analysis breaks down ROE into three components: profitability, efficiency, and leverage. Mathematically, the DuPont Analysis uses the following formula:

$$\text{ROE} = \text{Net Profit Margin} \times \text{Asset Turnover} \times \text{Equity Multiplier}$$

$$= \frac{\text{Net Profit}}{\text{Sales}} \times \frac{\text{Sales}}{\text{Assets}} \times \frac{\text{Assets}}{\text{Equity}}$$

The focus is not on the ROE itself but on how it is achieved via these three components. ROE may go up, but it may be due to additional leverage instead of superior margin or improved efficiency. Applying the formula to Nestlé will show the following:

DuPont ROE Analysis		20X7	20X6
Net Profit Margin	$= \frac{\text{Net Profit}}{\text{Sales}}$	$\frac{8,883}{89,449} = 9.9\%$	$\frac{9,467}{88,785} = 10.7\%$
Asset Turnover	$= \frac{\text{Sales}}{\text{Assets}}$	$\frac{89,449}{131,901} = 67.8\%$	$\frac{88,785}{123,992} = 71.6\%$
Equity Multiplier	$= \frac{\text{Assets}}{\text{Equity}}$	$\frac{131,901}{65,981} = 199.9\%$	$\frac{123,992}{63,986} = 193.8\%$
		DuPont ROE = 13.5%	DuPont ROE = 14.8%

We can see that ROE has gone down because of decrease in profitability and efficiency, despite the increase in leverage.

Investment Ratios

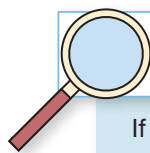
Investors buy shares to earn a return on their investment. They pay close attention to how much each share they own is generating profit, cash, and dividends, and the market price for each share. Let's look at some common ratios that can help you decide if your investment is performing well.

Earnings per Ordinary Share. Earnings per ordinary share, or simply earnings per share (EPS), is the amount of net income earned for each outstanding ordinary share (see Chapter 10). EPS is probably the most widely quoted of all financial statistics. It's also the only ratio that appears at the bottom of the Income Statement (required by *IAS 1—Presentation of Financial Statements*) and the only financial ratio that has an accounting standard (*IAS 33—Earnings per share*).

Earnings per share is computed by dividing net income available to ordinary shareholders by the weighted-average number of ordinary shares outstanding during the year. Preference dividends are subtracted from net income because the preference shareholders have a prior claim to their dividends. Ikon has 10,000 ordinary shares outstanding and no preference shares (and thus has no preference dividends). The firm's EPS ratios for 20X7 and 20X6 can be calculated as:

		<i>Ikon's Earnings per Ordinary Share</i>	
Formula		20X7	20X6
Earnings per share of share capital	$= \frac{\text{Net income} - \text{Preference dividends}}{\text{Weighted-average number of ordinary shares outstanding}}$	$\frac{\$48,000 - \$0}{10,000} = \$4.80$	$\frac{\$26,000 - \$0}{10,000} = \$2.60$

Ikon Furniture's EPS increased 85% during 20X7, and that's good news. The Ikon shareholders should not expect such a significant boost every year. Most companies can only expect to increase their EPS by a few percentage points annually.



A Closer Look

If you pay closer attention to the denominator in the EPS formula, you will notice that it is labeled "weighted-average number of ordinary shares outstanding." This means we are not using a simple arithmetic average but need to take into account how long a certain number of shares have been outstanding. For example, if you started the year on January 1, 20X6, with 10,000 shares and on November 1, 20X6, issued an additional 6,000 shares (total 16,000), the weighted average would be 11,000 (i.e., $10,000 \times 10/12$ months) + $(16,000 \times 2/12)$ months) shares for the period. For listed companies, there may be other equity transactions such as stock dividends, stock splits, and purchase of treasury shares that change the weighted-average of ordinary stock outstanding.

In addition, *IAS 1* and *IAS 33* require two versions of EPS: basic and diluted earnings per share. The latter takes into account potentially dilutive financial instruments such as convertible bonds, convertible preference shares, and stock options. This is beyond the scope of this text.

Price/Earnings Ratio. The **price/earnings ratio** shows how much an investor is willing to pay for each unit of earnings. This ratio, abbreviated P/E, appears in every financial section of newspapers and online financial databases. The P/E ratio makes use of the **earnings per share (EPS)** ratio we calculated earlier.

Calculations for the P/E ratios of Ikon Furniture, Inc., follow. Suppose the market price of Ikon's ordinary share was \$60 at the end of 20X7 and \$35 at the end of 20X6. Share prices can be obtained from a company's website, a financial publication or website, or a stockbroker.

Formula	Ikon's Price/Earnings Ratio	
	20X7	20X6
P/E ratio = $\frac{\text{Market price per ordinary share}}{\text{Earnings per share}}$	$\frac{\$60.00}{\$4.80} = 12.5$	$\frac{\$35.00}{\$2.60} = 13.5$

Given Ikon Furniture's 20X7 P/E ratio of 12.5, we would say that the company's share is selling at 12.5 times earnings. Each \$1 of Ikon's earnings is worth \$12.50 to the stock market. The P/E ratio reflects the market's overall expectation of a company's performance. The more optimistic the market is, the higher the P/E ratio. The home furnishing average PE ratio is about 14.3 times earnings.

Dividend Yield. **Dividend yield** is the ratio of dividends per share of stock to the share's market price. This ratio measures the percentage of a share's market value returned annually to the shareholders as dividends. For investors looking for a steady stream of dividend payouts, dividend yield is a very important ratio to consider in their investment decisions.

Ikon Furniture paid annual cash dividends of \$1.20 per share in 20X7 and \$1.00 in 20X6. The market prices of the company's ordinary share were \$60 in 20X7 and \$35 in 20X6. The firm's dividend yields on ordinary share are:

		<i>Dividend Yield on Ikon's Share Capital</i>	
Formula		20X7	20X6
Dividend yield on ordinary share*	$\frac{\text{Dividend perordinary share}}{\text{Market price perordinary share}}$	$\frac{\$1.20}{\$60.00} = 2.0\%$	$\frac{\$1.00}{\$35.00} = 2.9\%$

*Dividend yields may also be calculated for preference shares.

Preference shareholders pay special attention to this ratio because they invest primarily to receive dividends. The same formula can be adjusted to calculate dividend yield only for preference shareholders: dividend per preference share divided by market price per preference share.

An investor who buys Ikon Furniture ordinary shares for \$60 can expect to receive around 2% of the investment annually in the form of cash dividends. Dividend yields vary widely, from 5% to 8% for mature, established firms (such as P&G and Unilever) down to the range of 0% to 3% for younger, growth-oriented companies.

Book Value per Ordinary Share. Book value per ordinary share is simply ordinary shareholders' equity divided by the number of ordinary shares outstanding. Ordinary equity equals total equity less preference equity.

Ikon Furniture has no preference shares outstanding. Calculations of its book value per ordinary share follow. Recall that 10,000 ordinary shares were outstanding.

		<i>Book Value per Share of Ikon's Share Capital</i>	
Formula		20X7	20X6
Book value per ordinary share	$= \frac{\text{Totalshareholders' equity} - \text{Preferenceequity}}{\text{Number ofordinary sharesoutstanding}}$	$\frac{\$356,000 - \$0}{10,000} = \$35.60$	$\frac{\$320,000 - \$0}{10,000} = \$32.00$

Book value indicates the recorded accounting amount for each ordinary share outstanding. Many experts believe book value is not useful for investment analysis because it bears no relationship to market value and provides little information beyond what's reported on the Balance Sheet. But some investors do base their investment decisions on book value. Specifically, they would rank shares by the ratio of market price to book value. The lower the ratio, the more attractive the share. These investors are called "value" investors, as contrasted with "growth" investors, who focus more on trends in net income. The home furnishing industry's price to book value is about 1.4 times.

Investors often extend this ratio by comparing it to the current price for each share. For example, if the book value per share is \$2 and the current market price is \$6, we can say the "price to book" is 3 times.

Putting it All Together

What does the outlook for the future look like for Ikon Furniture? If the company can stay on the same path it has followed for the past two years, it looks bright. It appears that its earnings per share are solid, and its margins, ROA, and ROE ratios are all above average for its industry. From the standpoint of liquidity and leverage, it also appears to be in good shape, with higher liquidity,

excellent debt and interest coverage, and lower debt ratios than its industry. The company’s P/E ratio of 12:1 is below the industry average, and it pays a 2% dividend. All of these factors make Ikon Furniture shares look like a good investment.

CASH
FLOW

Using the Statement of Cash Flows

You may have noticed that the ratios we described earlier are all based on Income Statement and Balance Sheet. What about cash flows? We covered Free Cash Flow and Cash Realization Ratio earlier (in Chapter 11), but in general, cash flow ratios have received less attention and coverage than they deserve. Fortunately, they are not a whole new set of ratios! Cash flow ratios are usually alternative versions of the ratios we discussed earlier, with the emphasis changed to cash from operations (CFO). Here are some of the cash flow ratios you could use:

Standard ratios discussed earlier	Cash flow version	Variation from standard ratios
Current ratio	CFO coverage ratio	$\text{CFO} \div \text{current liabilities}$
Debt ratio	CFO solvency ratio	$\text{CFO} \div \text{total liabilities}$
Net profit margin	CFO margin ratio	$\text{CFO} \div \text{sales revenue}$
Times-interest-earned	CFO interest ratio	$(\text{CFO} + \text{interest paid}) \div \text{interest paid}$

USE OTHER INFORMATION TO MAKE
INVESTMENT DECISIONS

Economic Value Added (EVA®)

4
Use other information to make investment decisions

The top managers of Coca-Cola, Quaker Oats, and other leading companies use **economic value added (EVA®)** to evaluate operating performance. EVA® combines accounting and finance to measure whether operations have increased shareholder wealth. EVA® can be computed as follows:

$$\text{EVA}^{\text{®}} = \text{Net income} + \text{Interest expense} - \text{Capital charge}$$

$$\text{Capital charge} = \left(\begin{matrix} \text{Beginning balances} \\ \text{Notes payable} + \text{Current maturities of long-term debt} + \text{Long-term debt} + \text{Shareholders' equity} \end{matrix} \right) \times \text{Cost of capital}$$

All amounts for the EVA® computation, except the **cost of capital**, come from the financial statements. The EVA model uses net income, but adjusted for certain expenses (such as non-cash expenses and even marketing expenses). For simplicity, we shall use net income as a starting point and make some simple adjustments. The cost of capital is a weighted average of the returns demanded by the company’s shareholders and lenders. Cost of capital varies with the company’s level of risk. For example, shareholders would demand a higher return from a start-up company than from Nestlé because the new company is untested and, therefore, more risky. Lenders would also charge the new company a higher interest rate because of its greater risk. Thus, the new company has a higher cost of capital than Nestlé.

The cost of capital is a major topic in finance classes. In the following discussions we assume a value for the cost of capital (such as 10%, 12%, or 15%) to illustrate the computation of EVA®.

The idea behind EVA[®] is that the returns to the company's shareholders (net income) and to its creditors (interest expense) should exceed the company's **capital charge**. The capital charge is the amount that shareholders and lenders charge a company for the use of their money. A positive EVA[®] amount suggests an increase in shareholder wealth, and so the company's shares should remain attractive to investors. If EVA[®] is negative, shareholders will probably be unhappy with the company and sell its shares, resulting in a decrease in the share price. Different companies tailor the EVA[®] computation to meet their own needs.

Let's apply EVA[®] to Nestlé. For the income measure, we shall use net profit but add back impairment losses and depreciation, i.e., CHF 8,883 + 3,772 + 1,736 = CHF 14,391 million. We can continue to calculate Nestlé's EVA[®] as follows, assuming a 10% cost of capital.

$$\begin{aligned}
 \text{Nestlé's EVA}^{\text{®}} &= \text{Adjusted net income} + \text{Interest expense} - \left(\begin{array}{c} \text{(Beginning balances)} \\ \text{Short-term borrowings} + \text{Long-term debt} + \text{Shareholders' equity} \end{array} \right) \times \text{Cost of capital} \\
 &= \frac{14,391 + 758}{15,149} - \left(\frac{9,629 + 11,601 + 63,986}{85,216} \right) \times 0.10 \\
 &= \frac{15,149}{15,149} - \frac{8,522}{8,522} \\
 &= \frac{6,627}{6,627}
 \end{aligned}$$

By this measure, Nestlé's operations added CHF 6,627 million of value to its shareholders' wealth after meeting the company's capital charge. This performance is considered very strong, especially given the tough economic conditions around the world.

OTHER ISSUES IN FINANCIAL STATEMENT ANALYSIS

Limitations of Ratio Analysis

Business decisions are made in a world of uncertainty. As useful as financial ratios are, they aren't a cure-all. Consider a physician's use of a thermometer. A reading of 40° Celsius (or 104.0° Fahrenheit) tells a doctor something is wrong with the patient, but that doesn't indicate what the problem is or how to cure it.

In financial analysis, a sudden drop in any ratio may signal that something is wrong, but that doesn't identify the problem. A manager must analyze the figures to learn what caused the ratio to fall. The manager must evaluate all the ratios in the light of factors such as increased competition or a slowdown in the economy. Business models and strategies would also help explain the financial ratios you calculate. For example, you would expect a company selling luxury goods to have higher profit margins than another selling household goods. Sales turnover for the latter, however, would be significantly higher than for the former.

Legislation, international affairs, scandals, and other factors can turn profits into losses. To be useful, ratios should be analyzed over a period of years to consider all relevant factors. Any one year, or even any two years, may not represent the company's performance over the long term.

You must also remember that the numbers on the financial statement are a result of the company's applications of accounting standards that require significant judgment and often rely on estimates and assumptions. There may also be differences in accounting standards in one region versus other. For example, Sinopec (China's largest oil company) financial statements are prepared under Chinese Accounting Standards (CAS) and Tata Motors' under Indian GAAP. For introductory accounting courses, we may assume that the differences are minimal. Analyzing the numbers without consideration of the accounting policies that resulted in those numbers is like the old adage, "garbage in, garbage out."

Red Flags in Financial Statement Analysis

Recent accounting scandals have highlighted the importance of *red flags* in financial statement analysis. The following conditions may mean a company is very risky:

- **Earnings Problems.** Have income from continuing operations and net income decreased for several years in a row? Has income turned into a loss? This may be okay for a company in a cyclical industry, such as an airline or a home builder, but companies usually cannot survive losses in consecutive years.
- **Decreased Cash Flow.** Cash flow validates earnings. Is cash flow from operations consistently lower than net income? Are the sales of PPE a major source of cash? If so, the company may be facing a cash shortage.
- **Too Much Debt.** How does the company's debt ratio compare to that of major competitors and to the industry average? If the debt ratio is much higher than average, the company may be unable to pay debts during tough times.
- **Inability to Collect Receivables.** Are receivables resident periods growing faster than for other companies in the industry? A cash shortage may be looming.
- **Buildup of Inventories.** Is inventory turnover slowing down? If so, the company may be unable to move products, or it may be overstating inventory as reported on the Balance Sheet. Recall from the cost-of-goods-sold model that one of the easiest ways to overstate net income is to overstate ending inventory.
- **Trends of Sales, Inventory, and Receivables.** Sales, receivables, and inventory generally move together. Increased sales lead to higher receivables and require more inventory in order to meet demand. Strange movements among these items may spell trouble.

The Decision Guidelines feature summarizes the most widely used ratios.



DECISION GUIDELINES

USING RATIOS IN FINANCIAL STATEMENT ANALYSIS

Milla and Ryssa operate a financial services firm. The two sisters manage other people's money and do most of their own financial statement analysis. Suppose they are analyzing a company on behalf of their clients and want to find out the company's efficiency, financial strength, performance, and suitability as an investment. They use the standard ratios we have covered in this chapter.

Ratio	Computation	Information Provided
Measure efficiency of a company's operations:		
1. Inventory turnover	$\frac{\text{Cost of goods sold}}{\text{Average inventory}}$	Indicates the salability of inventory—the number of times a company sells its average level of inventory during a year. Can be expressed in days as inventory resident period.
2. Accounts receivable turnover	$\frac{\text{Sales revenue credit sales}}{\text{Average net accounts receivable}}$	Measures the ability to collect cash from customers. Can be expressed in days as receivables collection period.
3. Payable turnover	$\frac{\text{Cost of goods sold}}{\text{Average accounts payable}}$	Measures the frequency of payments to trade creditors. Can be expressed in days as payable payment period.

Ratio	Computation	Information Provided
4. Cash conversion cycle	Receivable collection period + Inventory resident period – Payable outstanding period	Indicates the speed at which an entity is able to convert cash from its inventory and receivables, less the time it takes to settle accounts payable.
5. Asset turnover	$\frac{\text{Sales}}{\text{Average total assets}}$	Measures the ability to use assets to generate revenue.
Measure financial strengths (short-term and long-term):		
6. Current ratio	$\frac{\text{Current assets}}{\text{Current liabilities}}$	Measures the ability to pay current liabilities with current assets.
7. Quick ratio	$\frac{\text{Sales}}{\text{Average total assets}}$	Shows the ability to pay all current liabilities if they come due immediately. Sometimes calculated as (current assets – inventory) divided by current liabilities.
8. Debt ratio	$\frac{\text{Total liabilities}}{\text{Total assets}}$	Indicates percentage of assets financed with liabilities or debt.
9. Times-interest-earned ratio	$\frac{\text{Income from operations}}{\text{Interest expense}}$	Measures the number of times operating income can cover interest expense. Also called interest coverage ratio.
Measure profitability:		
10. Gross/Operating/Net profit margin	$\frac{\text{Gross or Operating or Net profit}}{\text{Net sales}}$	Shows the percentage of each sales dollar earned as income. Variants include gross profit, operating profit, and net profit.
11. Return on total assets (ROA)	$\frac{\text{Net income}}{\text{Average total assets}}$	Measures how profitably a company uses its assets. For the numerator, alternative formulas exist (net income plus interest expense, operating profit, or earnings before interest and tax).
12. Return on ordinary shareholders' equity (ROE)	$\frac{\text{Net income} - \text{Preference dividends}}{\text{Average ordinary shareholders' equity}}$	Gauges how much income is earned for on ordinary shareholder equity.
Analyze shares as potential investment:		
13. Earnings per ordinary share (EPS)	$\frac{\text{Net income} - \text{Preference dividends}}{\text{Weighted-average number of ordinary shares outstanding}}$	The amount of net income on a per-ordinary-share basis.
14. Price/earnings (P/E) ratio	$\frac{\text{Market price per ordinary share}}{\text{Earnings per share}}$	Indicates the market price for one currency unit of earnings.
15. Dividend yield	$\frac{\text{Dividend per ordinary share (or preference)}}{\text{Market price per ordinary share (or preference)}}$	Shows the percentage of a share's market value returned as dividends to shareholders in each period.
16. Book value per ordinary share	$\frac{\text{Total shareholders' equity} - \text{Preference equity}}{\text{Number of ordinary shares outstanding}}$	Indicates the recorded accounting amount for each ordinary share outstanding.

The following financial data are adapted from the annual reports of Artisan Corporation:

A1					
	A	B	C	D	E
	Artisan Corporation				
	Four-Year Selected Financial Data				
1	Years Ended January 31, 20X6, 20X5, 20X4, and 20X3				
2	Operating Results*	20X6	20X5	20X4	20X3
3	Net Sales	\$ 13,848	\$ 13,673	\$ 11,635	\$ 9,054
4	Cost of goods sold and occupancy expenses				
5	excluding depreciation and amortization	9,704	8,599	6,775	5,318
6	Interest expense	109	75	45	46
7	Income from operations	338	1,445	1,817	1,333
8	Net earnings (net loss)	(8)	877	1,127	824
9	Cash dividends	76	75	76	77
10					
11	Financial Position				
12	Merchandise inventory	1,677	1,904	1,462	1,056
13	Total assets	7,591	7,012	5,189	3,963
14	Current ratio	1.48:1	0.95:1	1.25:1	1.20:1
15	Shareholders' equity	3,010	2,928	2,630	1,574
16	Average number of ordinary shares outstanding				
17	(in thousands)	860	879	895	576
18					

*Dollar amounts are in thousands.

Requirement

1. Compute the following ratios for 20X4 through 20X6, and evaluate Artisan's operating results. Are operating results strong or weak? Did they improve or deteriorate during the three-year period? Your analysis will reveal a clear trend.
 - a. Gross profit margin
 - b. Net profit margin
 - c. Earnings per share
 - d. Inventory turnover
 - e. Times-interest-earned ratio
 - f. Return on shareholders' equity

Answer

	20X6	20X5	20X4
1. Gross profit margin	$\frac{\$13,848 - \$9,704}{\$13,848} = 29.9\%$	$\frac{\$13,673 - \$8,599}{\$13,673} = 37.1\%$	$\frac{\$11,635 - \$6,775}{\$11,635} = 41.8\%$
2. Net profit margin	$\frac{\$(8)}{\$13,848} = (0.06)\%$	$\frac{\$877}{\$13,673} = 6.4\%$	$\frac{\$1,127}{\$11,635} = 9.7\%$
3. Earnings per share	$\frac{\$(8)}{860} = \(0.01)	$\frac{\$877}{879} = \1.00	$\frac{\$1,127}{895} = \1.26
4. Inventory turnover	$\frac{\$9,704}{(\$1,677 + \$1,904)/2} = 5.4 \text{ times}$	$\frac{\$8,599}{(\$1,904 + \$1,462)/2} = 5.1 \text{ times}$	$\frac{\$6,775}{(\$1,462 + \$1,056)/2} = 5.4 \text{ times}$
5. Times-interest-earned ratio	$\frac{\$338}{\$109} = 3.1 \text{ times}$	$\frac{\$1,445}{\$75} = 19.3 \text{ times}$	$\frac{\$1,817}{\$45} = 40.4 \text{ times}$
6. Return on equity	$\frac{\$(8)}{(\$3,010 + \$2,928)/2} = (0.3\%)$	$\frac{\$877}{(\$2,928 + \$2,630)/2} = 31.6\%$	$\frac{\$1,127}{(\$2,630 + \$1,574)/2} = 53.6\%$

Evaluation: During this period, Artisan's operating results deteriorated on all these measures except inventory turnover. The gross profit percentage is down sharply, as are the times-interest-earned ratio and all the return measures. From these data it is clear that Artisan could sell its merchandise, but not at the mark-ups the company enjoyed in the past. The final result, in 20X6, was a net loss for the year.

REVIEW | Financial Statement Analysis

Quick Check (Answers are given at the end of the chapter.)

Analyze the Odyssey Company's financial statements by answering the questions that follow. Odyssey owns a chain of restaurants.

A1			
	A	B	C
	Oullette Company Consolidated Statements of Income (Adapted) Years Ended December 31, 20X7 and 20X6		
1			
2	(In thousands, except per share data)	20X7	20X6
3	Revenues:		
4	Sales by Company-operated restaurants	\$ 13,200	\$ 11,100
5	Revenues from franchised and affiliated restaurants	4,500	3,700
6	Total revenues	17,700	14,800
7	Expenses:		
8	Cost of goods sold	3,300	3,108
9	Payroll and employee benefits	3,200	3,000
10	Occupancy and other operating expenses	2,900	2,800
11	Franchised restaurants—occupancy expenses	949	850
12	Selling, general, and administrative expenses	1,820	1,730
13	Other operating expense, net	510	855
14	Total operating expenses	12,679	12,343
15	Operating income	5,021	2,457
16	Interest expense	370	345
17	Other non-operating expense, net	140	168
18	Income before income taxes	4,511	1,944
19	Income tax expense	1,820	820
20	Net income	\$ 2,691	\$ 1,124
21	Per-ordinary-share basics:		
22	Net income	\$ 2.69	\$ 1.15
23	Dividends per ordinary share	\$ 0.50	\$ 0.24
24			

A1			
	A	B	C
1	Oullette Company		
2	Consolidated Balance Sheets		
3	December 31, 20X7 and 20X6		
4	(In thousands, except per share data)	20X7	20X6
5	Assets		
6	Current Assets		
7	Cash and equivalents	\$ 690	\$ 455
8	Accounts and notes receivable	780	840
9	Inventories	140	120
10	Prepaid expense and other current assets	580	440
11	Total current assets	2,190	1,855
12	Other Assets		
13	Investments in affiliates	1,150	1,055
14	Goodwill, net	1,780	1,590
15	Miscellaneous	990	1,100
16	Total other assets	3,920	3,745
17	Property and Equipment		
18	Property and equipment, at cost	28,800	26,500
19	Accumulated depreciation and amortization	(8,850)	(7,900)
20	Net property and equipment	19,950	18,600
21	Total assets	\$ 26,060	\$ 24,200
22	Liabilities and Shareholders' Equity		
23	Current liabilities		
24	Accounts payable	\$ 520	\$ 675
25	Income taxes	70	14
26	Other taxes	230	180
27	Accrued interest	189	196
28	Accrued restructuring and restaurant closing costs	110	385
29	Accrued payroll and other liabilities	890	795
30	Current maturities of long-term debt	365	305
31	Total current liabilities	2,374	2,550
32	Long-term debt	8,700	9,500
33	Other long-term liabilities and minority interests	690	520
34	Deferred income taxes	1,005	1,015
35	Shareholders' Equity		
36	Preference shares, no par value; authorized—140.0 million shares;		
37	issued—none	—	—
38	Ordinary shares, \$0.01 par value; authorized—2.0 billion shares;		
39	issued—1,400 million shares	14	14
40	Additional paid-in capital	1,786	1,662
41	Unearned ESOP compensation	(85)	(101)
42	Retained earnings	21,741	19,550
43	Accumulated other comprehensive income (loss)	(815)	(1,570)
44	Ordinary shares in treasury, at cost; 400 and 420 million shares	(9,350)	(8,940)
45	Total shareholders' equity	13,291	10,615
46	Total liabilities and shareholders' equity	\$ 26,060	\$ 24,200

1. Horizontal analysis of Odyssey's Income Statement for 20X7 would show which of the following for Selling, General, and Administrative expenses?
 - a. 1.05
 - b. 0.95
 - c. 0.68
 - d. None of the above
2. Vertical analysis of Odyssey's Income Statement for 20X7 would show which of the following for Selling, General, and Administrative expenses?
 - a. 0.144
 - b. 0.138
 - c. 0.103
 - d. None of the above
3. Which item on Odyssey's income statement has the most favorable trend during 20X6–20X7?
 - a. Total revenues
 - b. Food and paper costs
 - c. Net income
 - d. Payroll and employee benefits
4. On Odyssey's common-size Balance Sheet, Goodwill would appear as
 - a. up by 11.9%.
 - b. \$1,780 million.
 - c. 10.06% of total revenues.
 - d. 0.068.
5. A good benchmark for Odyssey Company would be
 - a. Volvo.
 - b. Whataburger.
 - c. Microsoft.
 - d. All of the above.
6. Odyssey's inventory turnover for 20X7 was
 - a. 61 times.
 - b. 25 times.
 - c. 17 times.
 - d. 72 times.
7. Odyssey's acid-test ratio at the end of 20X7 was
 - a. 0.06.
 - b. 2.83.
 - c. 0.92.
 - d. 0.62.
8. Odyssey's average collection period for accounts and notes receivables is
 - a. 17 days.
 - b. 32 days.
 - c. 15 days.
 - d. 2 days.
9. The average debt ratio for most companies is 0.64. Odyssey's total debt position looks
 - a. risky.
 - b. safe.
 - c. middle-ground.
 - d. cannot tell from the financials.
10. Odyssey's return on total revenues for 20X7 was
 - a. \$2.69.
 - b. \$1.16.
 - c. 10.33%.
 - d. 15.2%.
11. Odyssey's return on shareholders' equity for 20X7 was
 - a. 10.3%.
 - b. 15.2%.
 - c. 22.5%.
 - d. \$2,691 million.
12. On May 31, 20X7, Odyssey's ordinary shares sold for \$30 per share. At that price, how much did investors say \$1 of the company's net income was worth?
 - a. \$30.00
 - b. \$1.00
 - c. \$10.99
 - d. \$11.15
13. On May 31, 20X7, Odyssey's ordinary shares sold for \$30 per share and dividends per share were \$0.50. Compute Odyssey's dividend yield during 20X7.
 - a. 4.1%
 - b. 2.9%
 - c. 5.0%
 - d. 1.7%

14. How much EVA® did Odyssey generate for investors during 20X7? Assume the cost of capital was 5% and use unadjusted net income.
- | | |
|--------------------|--------------------|
| a. \$2.045 million | c. \$3.061 million |
| b. \$1.943 million | d. \$2.691 million |

Accounting Vocabulary

asset turnover ratio (p. 719) An indicator of the efficiency with which a company is deploying its assets in generating revenue.

benchmarking (p. 711) The comparison of an entity to another entity (competitor or peer) or against industry averages, with the objective of providing context to the evaluation of the entity.

book value per ordinary share (p. 727) Ordinary shareholders' equity divided by the number of ordinary shares outstanding. The recorded amount for each ordinary share outstanding.

capital charge (p. 729) The amount that shareholders and lenders charge a company for the use of their money. Calculated as beginning balances of (Notes payable + Loans payable + Long-term debt + Shareholders' equity) $\times$ Cost of capital.

cash conversion cycle (p. 717) The length of time it takes a company to convert cash from its inventory purchases and receivables. Calculated as inventory resident period plus receivable collection period less payable outstanding period.

common-size financial statement (p. 712) A financial statement that reports only percentages (no dollar amounts).

cost of capital (p. 728) A weighted average of the returns demanded by the company's shareholders and lenders.

current ratio (p. 719) Current assets divided by current liabilities. Measures a company's ability to pay current liabilities with current assets.

debt ratio (p. 721) Ratio of total liabilities to total assets. States the proportion of a company's assets that is financed with debt.

debt-to-equity (p. 721) A measure of indebtedness; the amount of liabilities (or interest-bearing liabilities) as a proportion of total equity.

dividend yield (p. 726) Ratio of dividends per share to the share's market price. Tells the percentage of a share's market value that the company returns to shareholders as dividends.

earnings per share (EPS) (p. 726) Amount of a company's net income earned for each ordinary share outstanding.

economic value added (EVA®) (p. 728) Used to evaluate a company's operating performance. EVA combines the concepts of accounting income and corporate finance to measure whether the company's operations have increased shareholder wealth. $EVA = \text{Net income} + \text{Interest expense} - \text{Capital charge}$.

gearing (p. 721) A measure of indebtedness of an entity, also known as leverage, typically measured by the proportion of assets financed by liabilities (or alternatively only interest-bearing liabilities).

gross profit margin (p. 723) The amount of gross profit earned from each currency unit of sales.

horizontal analysis (p. 706) Study of percentage changes in comparative financial statements.

interest-coverage ratio (p. 721) Another name for *times-interest-earned ratio*.

inventory resident period (p. 716) The average length of time (in days) to sell inventory, based on *inventory turnover*. Part of *cash conversion cycle*.

inventory turnover (p. 716) Ratio of cost of goods sold to average inventory. Indicates how rapidly inventory is sold. Can also be expressed in days (*inventory resident period*).

leverage (p. 721) The degree of external financing of an entity. Earning more income on borrowed money than the related interest expense, thereby increasing the earnings for the owners of the business.

liquidity ratios (p. 719) Financial strength ratios that focus on short-term obligations, typically current liabilities.

net profit margin (p. 723) The amount of net profit earned from each currency unit of sales. Also called *return on sales*.

operating profit margin (p. 723) The amount of operating profit earned from each currency unit of sales.

payable outstanding period (p. 717) The average length of time (in days) to pay account payables, based on *payable turnover*. Part of *cash conversion cycle*.

payable turnover (p. 717) Ratio of cost of goods sold to average payables. Indicates how quickly trade creditors are paid. Can also be expressed in days (payables outstanding period).

price/earnings ratio (p. 726) Ratio of the market price of an ordinary share to the company's earnings per share. Measures the value that the share market places on one currency unit of a company's earnings.

quick ratio (p. 720) Ratio of the sum of cash plus short-term investments plus net current receivables to total current liabilities. Tells whether the entity can pay all its current liabilities if they come due immediately. Also called the *acid-test ratio*.

receivable collection period (p. 717) The average length of time (in days) to collect receivables, based on *receivable turnover*. Part of *cash conversion cycle*.

return on ordinary shareholders' equity (p. 723) Net income minus preference dividends, divided by average ordinary shareholders' equity. A measure of profitability. Also called *return on equity (ROE)*.

return on sales (p. 723) Another name for *net profit margin*.

return on total assets (p. 723) Net income divided by average total assets. This ratio measures a company's success in using its assets to earn income for the persons who finance the business. Also called *return on assets (ROA)*.

solvency ratios (p. 719) Financial strength ratios that focus on both short- and long-term obligations.

times-interest-earned ratio (p. 721) Ratio of income from operations (or earnings before interest and tax) to interest expense. Measures the number of times that operating income

can cover interest expense. Also called the *interest-coverage ratio*.

trend percentages (p. 709) A form of horizontal analysis that indicates the direction a business is taking.

vertical analysis (p. 710) Analysis of a financial statement that reveals the relationship of each statement item to a specified base, which is the 100% figure.

working capital (p. 719) Current assets minus current liabilities; measures a business's ability to meet its short-term obligations with its current assets.

ASSESS YOUR PROGRESS

Short Exercises

S12-1. (Learning Objective 1: Performing horizontal analysis of revenues and net income)

Fitzgerald Corporation reported the following amounts on its 20X6 comparative Income Statement:

L0 1

(In thousands)	20X6	20X5	20X4
Revenues	\$10,727	\$9,832	\$9,411
Total expenses.....	6,325	6,099	5,772

Perform a horizontal analysis of revenues and net income—both in dollar amounts and in percentages—for 20X6 and 20X5.

S12-2. (Learning Objective 1: Performing trend analysis of sales and net income) Fenton, Inc., reported the following sales and net income amounts:

L0 1

(In thousands)	20X6	20X5	20X4	20X3
Sales	\$12,010	\$9,680	\$8,470	\$8,240
Net income.....	635	515	390	310

Show Fenton's trend percentages for sales and net income. Use 20X3 as the base year.

S12-3. (Learning Objective 2: Performing vertical analysis to correct a cash shortage) Craft Software reported the following amounts on its Balance Sheets at December 31, 20X6, 20X5, and 20X4:

L0 2

(In thousands)	20X6	20X5	20X4
Cash.....	\$ 7,800	\$ 2,250	\$ 2,000
Receivables, net.....	36,400	22,500	24,000
Inventory.....	270,400	198,000	148,000
Prepaid expenses	10,400	18,000	12,000
Property, plant and equipment, net	195,000	209,250	214,000
Total assets	<u>\$520,000</u>	<u>\$450,000</u>	<u>\$400,000</u>

Sales and profits are high. Nevertheless, Craft is experiencing a cash shortage. Perform a vertical analysis of Craft Software's assets at the end of years 20X6, 20X5, and 20X4. Use the analysis to explain the reason for the cash shortage.

L0 3

S12-4. (Learning Objective 3: Comparing common-size Income Statements of two companies) Hartigan, Inc., and Pintal Corporation are competitors. Compare the two companies by converting their condensed Income Statements to common size.

(In millions)	Hartigan	Pintal
Net sales.....	\$10,850	\$8,700
Cost of goods sold.....	6,499	6,029
Selling and administrative expenses.....	3,125	1,688
Interest expense.....	54	35
Other expenses.....	33	43
Income tax expense.....	432	209
Net income.....	\$ 705	\$ 696

Which company earned more net income? Which company's net income was a higher percentage of its net sales? Explain your answer.

L0 4

S12-5. (Learning Objective 4: Evaluating the trend in a company's current ratio) Examine the financial data of Jacob Corporation.

Year Ended December 31	20X6	20X5	20X4
Operating Results			
Net income.....	\$ 220	\$ 120	\$ 119
Per ordinary share.....	\$1.23	\$0.93	\$0.63
Percent of sales.....	15.6%	17.6%	19.6%
Return on average shareholders' equity.....	14.0	17.0	20.0
Financial Position			
Current assets.....	\$ 560	\$ 455	\$ 445
Current liabilities.....	360	333	356
Working capital.....	200	112	89
Current ratio.....	1.56	1.37	1.25

Show how to compute Jacob's current ratio for each year 20X4 through 20X6. Is the company's ability to pay its current liabilities improving or deteriorating?

L0 4

S12-6. (Learning Objective 4: Evaluating a company's acid-test ratio) Use the Gargantua, Inc., Balance Sheet data below.

Requirements

1. Compute Gargantua, Inc.'s acid-test ratios at December 31, 20X6 and 20X5.
2. Use the comparative information from the table on the bottom of the following page for Horner, Inc., Isaacson Company, and Jona Companies Limited. Are Gargantua, Inc.'s acid-test ratios for 20X6 and 20X5 strong, average, or weak in comparison?

A1					
	A	B	C	D	E
1	Gargantua, Inc.				
2	Balance Sheets (Adapted)				
3				Increase (Decrease)	
4	(Dollar amounts in millions)	Dec 31, 20X6	Dec 31, 20X5	Amount	Percentage
5	Assets				
6	Current Assets				
7	Cash and cash equivalents	\$ 1,203	\$ 903	\$ 300	33.2%
8	Short-term investments	7	84	(77)	(91.7)
9	Receivables, net	246	256	(10)	(3.9)
10	Inventories	91	81	10	12.3
11	Prepaid expenses and other assets	203	343	(140)	(40.8)
12	Total current assets	1,750	1,667	83	5.0
13	Property, plant and equipment, net	3,619	3,396	223	6.6
14	Intangible assets	1,089	841	248	29.5
15	Other assets	824	718	106	14.8
16	Total assets	\$ 7,282	\$ 6,622	\$ 660	10.0%
17	Liabilities and Shareholders' Equity				
18	Current liabilities				
19	Accounts payable	\$ 977	\$ 884	\$ 93	10.50%
20	Income tax payable	39	69	(30)	(44)
21	Short-term debt	121	115	6	5.2
22	Other	70	73	(3)	(4)
23	Total current liabilities	1,207	1,141	66	5.8
24	Long-term debt	3,544	2,982	562	18.8
25	Other liabilities	1,177	1,046	131	12.5
26	Total liabilities	5,928	5,169	759	14.7
27	Shareholders' equity				
28	Share capital	—	—	—	—
29	Retained earnings	1,513	1,629	(116)	(7.1)
30	Accumulated other comprehensive (loss)	(159)	(176)	17	9.7
31	Total shareholders' equity	1,354	1,453	(99)	(6.8)
32	Total liabilities and shareholders' equity	\$ 7,282	\$ 6,622	\$ 660	10.0%

Company	Acid-Test Ratio
Horner, Inc. (Utility)	0.73
Isacson Company (Department store).....	0.68
Jona Companies Limited (Grocery store)	0.72

L0 4

S12-7. (Learning Objective 4: Computing and evaluating cash collection cycle) Use the Gargantua 20X6 Income Statement shown here and the Balance Sheet from Short Exercise 12-6 to compute its cash conversion cycle for the year 20X6. Identify all components of the cash conversion cycle clearly.

A1			
	A	B	C
	Gargantua, Inc.		
	Statements of Income (Adapted)		
1			
2	(Dollar amounts in millions)	Dec 31, 20X6	Dec 31, 20X5
3	Revenues	\$ 9,550	\$ 9,086
4	Expenses:		
5	Cost of goods sold	2,230	2,248
6	Payroll and employee benefits	2,134	2,004
7	Occupancy and other operating expenses	2,776	2,784
8	General and administrative expenses	1,175	1,131
9	Interest expense	154	136
10	Other expense (income), net	12	(32)
11	Income before income taxes	1,069	851
12	Income tax expense	288	252
13	Net income	\$ 781	\$ 599
14			

Do these measures look strong or weak? Give reasons for your answer.

L0 4

S12-8. (Learning Objective 4: Measuring ability to pay long-term debt) Use the financial statements of Gargantua, Inc., in Short Exercises 12-6 and 12-7.

Requirements

1. Compute the company's debt ratio at December 31, 20X6.
2. Compute the company's times-interest-earned ratio for 20X6. For operating income, use income before both interest expense and income taxes. You can simply add interest expense back to income before taxes.
3. Is Gargantua's ability to pay liabilities and interest expense strong or weak? Comment on the value of each ratio computed for questions 1 and 2.

L0 4

S12-9. (Learning Objective 4: Measuring profitability) Use the financial statements of Gargantua, Inc., in Short Exercises 12-6 and 12-7 to compute these profitability measures for 20X6. Show each computation.

- a. Gross, operating, and net profit margins.
- b. Return on total assets.
- c. Return on ordinary shareholders' equity.

L0 4

S12-10. (Learning Objective 4: Computing EPS and the price/earnings ratio) The annual report of Tri-State Cars, Inc., for the year ended December 31, 20X6, included the following items (in millions):

Preference share outstanding, 6%	\$425
Net income.....	\$510
Number of ordinary shares outstanding.....	100

Requirements

1. Compute earnings per share (EPS) and the price/earnings ratio for Tri-State Cars' shares. Round to the nearest cent. The price of a Tri-State Car share is \$57.12.
2. How much does the stock market say \$1 of Tri-State Cars' net income is worth?

S12-11. (Learning Objective 4: Using ratio data to reconstruct an Income Statement) A skeleton of Athol Country Florist's Income Statement appears as follows (amounts in thousands):

L0 4

A1		
	A	B
1	Income Statement	
2	Net sales	\$ 7,500
3	Cost of goods sold	(a)
4	Selling expenses	1,511
5	Administrative expenses	328
6	Interest expense	(b)
7	Other expenses	154
8	Income before taxes	1,046
9	Income tax expense	(c)
10	Net income	\$ (d)
11		

Use the following ratio data to complete Athol Country Florist's Income Statement:

- Inventory turnover was 4 (beginning inventory was \$748; ending inventory was \$726).
- Net profit margin is 12%.

S12-12. (Learning Objective 4: Using ratio data to reconstruct a Balance Sheet) A skeleton of Athol Country Florist's Balance Sheet appears as follows (amounts in thousands):

L0 4

A1				
	A	B	C	D
1	Balance Sheet			
2	Cash	\$ 85	Total current liabilities	\$ 1,900
3	Receivables	(a)	Long-term debt	(e)
4	Inventories	762	Other long-term liabilities	720
5	Prepaid expenses	(b)		
6	Total current assets	(c)		
7	PPE	(d)	Share capital	185
8	Other assets	2,100	Retained earnings	3,515
9	Total assets	\$ 7,400	Total liabilities and equity	\$ (f)
10				

Use the following ratio data to complete Athol Country Florist's Balance Sheet:

- Debt ratio is 0.50.
- Current ratio is 1.40.
- Acid-test ratio is 0.45.

S12-13. (Learning Objective 4: Analyzing a company based on its ratios) Take the role of an investment analyst at Standard Chartered. It is your job to recommend investments for your client. The only information you have is the following ratio values for two companies in the graphics software industry.

L0 4

Ratio	Graphit.net	Data Doctors
Days' sales in receivables.....	44	50
Inventory turnover	6	10
Gross profit percentage	69%	60%
Net income as a percent of sales.....	13%	14%
Times interest earned	17	11
Return on equity.....	36%	28%
Return on assets.....	15%	20%

Write a report to the Standard Chartered's investment committee. Recommend one company's shares over the other. State the reasons for your recommendation.

L0 5

S12-14. (Learning Objective 5: Measuring economic value added) Compute economic value added (EVA[®]) for Beverly Software. The company's cost of capital is 5%. Net income was \$780,000, interest expense \$408,000, beginning long-term debt \$720,000, and beginning shareholders' equity \$3,080,000. Use unadjusted net income in your calculations. Round all amounts to the nearest thousand dollars.

Should the company's shareholders be happy with the EVA[®]?

Exercises MyLab Accounting

Select A and B exercises can be found within MyLab Accounting, an online homework and practice environment. Your instructor may ask you to complete select exercises using MyLab Accounting.

Group A

L0 1

E12-15A. (Learning Objective 1: Computing year-to-year changes in working capital) What were the dollar amount of change and the percentage of each change in Wilderness Lodge's working capital during 20X6 and 20X5? Is this trend favorable or unfavorable?

	20X6	20X5	20X4
Total current assets	\$280,000	\$330,000	\$350,000
Total current liabilities	125,000	160,000	170,000

Which company appears to have the strongest financial position? Explain your reasoning.

L0 1

E12-16A. (Learning Objective 1: Performing horizontal analysis of an Income Statement) Prepare a horizontal analysis of the comparative Income Statements of Sensible Music Co. Round percentage changes to the nearest one-tenth percent (three decimal places).

■ spreadsheet

A1			
	A	B	C
	Sensible Music Co. Comparative Income Statements Years Ended December 31, 20X6 and 20X5		
1			
2		20X6	20X5
3	Total revenue	\$ 855,000	\$ 915,000
4	Expenses:		
5	Cost of goods sold	\$ 403,000	\$ 409,000
6	Selling and general expenses	233,000	262,000
7	Interest expense	9,300	10,600
8	Income tax expense	84,000	85,000
9	Total expenses	729,300	766,600
10	Net income	\$ 125,700	\$ 148,400
11			

L0 1

E12-17A. (Learning Objective 1: Computing trend percentages) Compute trend percentages for Palm Valley Sales & Service's total revenue and net income for the following five-year period, using year 0 as the base year. Round to the nearest full percent.

(In thousands)	Year 4	Year 3	Year 2	Year 1	Year 0
Total revenue	\$1,441	\$1,226	\$1,122	\$1,018	\$1,040
Net income	106	101	88	75	90

Which grew faster during the period, total revenue or net income?

E12-18A. (Learning Objective 2: Performing vertical analysis of a Balance Sheet) Fore Golf Company has requested that you perform a vertical analysis of its Balance Sheet to determine the component percentages of its assets, liabilities, and shareholders' equity.

L0 2

A1	A	B
	Fore Golf Company Balance Sheet December 31, 20X6	
1		
2	Assets	
3	Total current assets	\$ 41,230
4	Property, plant and equipment, net	112,290
5	Other assets	36,480
6	Total assets	\$ 190,000
7		
8	Liabilities	
9	Total current liabilities	\$ 46,930
10	Long-term debt	104,690
11	Total liabilities	125,700
12		
13	Shareholders' Equity	
14	Total shareholders' equity	38,380
15	Total liabilities and shareholders' equity	\$ 190,000
16		

E12-19A. (Learning Objective 3: Preparing a common-size Income Statement) Prepare a comparative common-size Income Statement for Sensible Music Co., using the 20X6 and 20X5 data of Exercise 12-16A and rounding to four decimal places.

L0 3

■ spreadsheet

E12-20A. (Learning Objective 3: Preparing a common-size Income Statement) Compare the 20X6 common-size Income Statement you performed earlier in E12-19A. How does this compare to Nestlé's vertical analysis in Exhibit 12-4? You may need to combine a few line items to make the common-size statements comparable.

L0 3

■ writing assignment

E12-21A. (Learning Objective 4: Computing five ratios) The financial statements of Smith News, Inc., include the following items:

L0 4

■ spreadsheet

	Current Year	Preceding Year
Balance Sheet:		
Cash	\$ 26,000	\$ 32,000
Short-term investments	14,000	20,000
Net receivables	50,000	73,000
Inventory	94,000	76,000
Prepaid expenses.....	9,000	8,000
Total current assets	193,000	209,000
Total current liabilities.....	129,000	96,000
Income Statement:		
Sales revenue	\$490,000	
Cost of goods sold	274,000	

Requirement

- Compute the following ratios for the current year:
 - Current ratio
 - Acid-test ratio
 - Receivable collection period
 - Inventory resident period
 - Payable outstanding period
 - Cash conversion cycle

LO 4

■ writing assignment

■ spreadsheet

E12-22A. (Learning Objective 4: Analyzing the ability to pay current liabilities) Barney Furniture Company has requested that you determine whether the company's ability to pay its current liabilities and long-term debts improved or deteriorated during 20X6. To answer this question, compute the following ratios for 20X6 and 20X5:

- a. Current ratio
- b. Acid-test ratio
- c. Debt ratio
- d. Times-interest-earned ratio

Summarize the results of your analysis in a written report.

	20X6	20X5
Cash	\$ 21,000	\$ 53,000
Short-term investments.....	32,000	15,000
Net receivables.....	117,000	127,000
Inventory.....	243,000	272,000
Prepaid expenses	18,000	4,000
Total assets.....	500,000	531,000
Total current liabilities	247,000	312,000
Long-term debt	27,000	134,000
Income from operations	191,000	160,000
Interest expense.....	39,000	45,000

LO 4

E12-23A. (Learning Objective 4: Analyzing profitability) Compute four ratios that measure the ability to earn profits for Harper Decor, Inc., whose comparative Income Statements follow:

A1			
	A	B	C
	Harper Decor, Inc.		
	Comparative Income Statements		
	Years Ended December 31, 20X6 and 20X5		
1			
2		20X6	20X5
3	Net sales	\$ 120,000	\$ 100,000
4	Cost of goods sold	64,000	52,000
5	Gross profit	56,000	48,000
6	Selling and general expenses	21,000	18,000
7	Income from operations	35,000	30,000
8	Interest expense	4,000	2,000
9	Income before income tax	31,000	28,000
10	Income tax expense	10,000	8,000
11	Net income	\$ 21,000	\$ 20,000
12			

Additional data:

	20X6	20X5	20X4
Total assets.....	\$104,000	\$100,000	\$83,000
Ordinary shareholders' equity	\$ 72,000	\$ 70,000	\$69,000
Preference dividends.....	\$ 3,000	\$ 2,000	\$ 1,000
Ordinary shares outstanding			
during the year	\$ 10,000	\$ 9,000	\$ 4,000

Did the company's operating performance improve or deteriorate during 20X6?

E12-24A. (Learning Objective 4: Evaluating a share as an investment) Evaluate the ordinary share of Regal Distributing Company as an investment. Specifically, use the three ordinary share ratios to determine whether the ordinary share increased or decreased in attractiveness during the past year.

L0 4**■ writing assignment**

	20X6	20X5
Net income.....	\$ 84,000	\$ 59,000
Dividends to ordinary shares.....	22,000	23,000
Total shareholders' equity at year-end..... (includes 90,000 ord. shares)	320,000	520,000
Preference shares, 5%	80,000	80,000
Market price per ordinary share at year-end.....	\$ 24.50	\$ 17.50

E12-25A. (Learning Objective 5: Using economic value added to measure corporate performance) Two companies with different economic-value-added (EVA[®]) profiles are Barton Oil Pipeline Incorporated and Crompton Bank Limited. Adapted versions of the two companies' financial statements are presented here (in millions):

L0 5

	Barton Oil Pipeline Inc.	Crompton Bank Limited
Balance Sheet data:		
Total assets	\$ 4,338	\$14,000
Interest-bearing debt	\$ 1,257	\$ 13
All other liabilities.....	2,675	2,605
Shareholders' equity	406	11,382
Total liabilities and equity.....	\$ 4,338	\$14,000
Income Statement data:		
Total revenue	\$11,000	\$ 3,800
Interest expense.....	75	6
Net income.....	\$ 190	\$ 1,230

Requirements

- Before performing any calculations, which company do you think represents the better investment? Give your reason.
- Compute the EVA[®] for each company and then decide which company's shares you would rather hold as an investment. Assume both companies' cost of capital is 8.5% and use unadjusted net income in your calculations.

Group B

E12-26B. (Learning Objective 1: Computing year-to-year changes in working capital)

L0 1

What were the euro amount of change and the percentage of each change in Ricardo Lodge's working capital during 20X6 and 20X5? Is this trend favorable or unfavorable?

	20X6	20X5	20X4
Total current assets	\$420,000	\$320,000	\$260,000
Total current liabilities	190,000	150,000	120,000

LO 1

■ writing assignment

E12-27B. (*Learning Objective 1: Performing horizontal analysis of an Income Statement*)

Prepare a horizontal analysis of the comparative Income Statements of Fashion Music Co. Round percentage changes to the nearest one-tenth percent (three decimal places).

A1			
	A	B	C
	Fashion Music Co.		
	Comparative Income Statements		
	Years Ended December 31, 20X6 and 20X5		
1			
2		20X6	20X5
3	Total revenue	€ 1,104,000	€ 919,000
4	Expenses:		
5	Cost of goods sold	€ 489,000	€ 400,450
6	Selling and general expenses	299,000	269,000
7	Interest expense	25,500	14,500
8	Income tax expense	107,500	86,850
9	Total expenses	921,000	770,800
10	Net income	€ 183,000	€ 148,200
11			

LO 1

E12-28B. (*Learning Objective 1: Computing trend percentages*) Compute trend percentages for Andover Valley Sales & Service's total revenue and net income for the following five-year period, using year 0 as the base year. Round to the nearest full percent.

(In thousands)	Year 4	Year 3	Year 2	Year 1	Year 0
Total revenue	€1,465	€1,279	€1,091	€1,030	€1,045
Net income	116	108	78	66	80

Which company appears to have the strongest financial position? Explain your reasoning.

LO 2

E12-29B. (*Learning Objective 2: Performing vertical analysis of a Balance Sheet*) Epsilon Golf Company has requested that you perform a vertical analysis of its Balance Sheet to determine the component percentages of its assets, liabilities, and shareholders' equity.

A1		
	A	B
	Epsilon Golf Company	
	Balance Sheet	
	December 31, 20X6	
1		
2	Assets	
3	Total current assets	\$ 46,000
4	Property, plant and equipment, net	212,000
5	Other assets	39,000
6	Total assets	\$ 297,000
7		
8	Liabilities	
9	Total current liabilities	\$ 54,000
10	Long-term debt	110,000
11	Total liabilities	164,000
12		
13	Shareholders' Equity	
14	Total shareholders' equity	133,000
15	Total liabilities and shareholders' equity	\$ 297,000
16		

E12-30B. (Learning Objective 3: Preparing a common-size Income Statement) Prepare a comparative common-size Income Statement for Fashion Music Co. using the 20X6 and 20X5 data of Exercise 12-27B and rounding to four decimal places.

L0 3
■ spreadsheet

E12-31B. (Learning Objective 3: Preparing a common-size Income Statement) Compare the year 20X6 common-size Income Statement you performed earlier in E12-21B. How does this compare to Nestlé's vertical analysis in Exhibit 12-4? You may need to combine a few line items to make the common-size statements comparable.

L0 3
■ writing assignment

E12-32B. (Learning Objective 4: Computing five ratios) The financial statements of Advent News, Inc., include the following items:

L0 4
■ spreadsheet

	Current Year	Preceding Year
Balance Sheet:		
Cash	\$ 65,000	\$ 91,000
Short-term investments	13,000	25,000
Net receivables	79,000	82,000
Inventory	93,000	75,000
Prepaid expenses.....	6,000	12,000
Total current assets	256	285,000
Total current liabilities.....	133	97,000
Income Statement:		
Sales revenue	\$498,000	
Cost of goods sold	278,000	

Requirement

1. Compute the following ratios for the current year:

- | | |
|---------------------------------|-------------------------------|
| a. Current ratio | d. Inventory resident period |
| b. Acid-test ratio | e. Payable outstanding period |
| c. Receivable collection period | f. Cash conversion cycle |

E12-33B. (Learning Objective 4: Analyzing the ability to pay current liabilities) Fred Furniture Company has requested that you determine whether the company's ability to pay its current liabilities and long-term debts improved or deteriorated during 20X6. To answer this question, compute the following ratios for 20X6 and 20X5. (Round your answers to two decimal places.)

- | | |
|--------------------|--------------------------------|
| a. Current ratio | c. Debt ratio |
| b. Acid-test ratio | d. Times-interest-earned ratio |

L0 4
■ writing assignment
■ spreadsheet

Summarize the results of your analysis in a written report.

	20X6	20X5
Cash	€ 27,000	€ 47,000
Short-term investments.....	33,000	4,000
Net receivables.....	120,000	135,000
Inventory.....	238,000	271,000
Prepaid expenses	22,000	8,000
Total assets.....	590,000	510,000
Total current liabilities	187,000	332,000
Long-term debt	147,000	84,000
Income from operations	191,000	169,000
Interest expense.....	41,000	43,000

L0 4

E12-34B. (Learning Objective 4: Analyzing profitability) Compute four ratios that measure the ability to earn profits for Collins Decor, Inc., whose comparative Income Statements follow:

A1			
	A	B	C
	Collins Decor, Inc.		
	Comparative Income Statements		
	Years Ended December 31, 20X6 and 20X5		
1			
2		20X6	20X5
3	Net sales	\$ 274,000	\$ 227,000
4	Cost of goods sold	135,000	116,000
5	Gross profit	149,000	111,000
6	Selling and general expenses	60,000	49,000
7	Income from operations	89,000	62,000
8	Interest expense	9,000	6,000
9	Income before income tax	80,000	56,000
10	Income tax expense	28,000	18,000
11	Net income	\$ 52,000	\$ 38,000
12			

Additional data:

	20X6	20X5	20X4
Total assets.....	\$249,000	\$239,000	\$227,000
Ordinary shareholders' equity	\$106,000	\$104,000	\$102,000
Preference dividends.....	\$ 17,000	\$ 15,000	\$ 13,000
Ordinary shares outstanding			
during the year	19,000	17,000	11,000

Did the company's operating performance improve or deteriorate during 20X6?

L0 4

■ **writing assignment**

E12-35B. (Learning Objectives 4: Evaluating a share as an investment) Evaluate the ordinary share of Basic Distributing Company as an investment. Specifically, use the three ordinary share ratios to determine whether the ordinary share increased or decreased in attractiveness during the past year.

	20X6	20X5
Net income.....	\$ 92,000	\$ 98,000
Dividends to ordinary shares.....	28,000	13,000
Total shareholders' equity at year-end.....	575,000	525,000
(includes 90,000 ord. shares)		
Preference shares, 5%	90,000	90,000
Market price per ordinary share		
at year-end.....	\$ 24.00	\$ 25.16

L0 5

E12-36B. (Learning Objective 5: Using economic value added to measure corporate performance) Two companies with different economic-value-added (EVA[®]) profiles are Houle Oil Pipeline, Inc., and Johnson Bank Limited. Adapted versions of the two companies' financial statements are presented here (in millions):

	Houle Oil Pipeline, Inc.	Johnson Bank Limited
Balance Sheet data:		
Total assets	\$ 4,338	\$14,451
Interest-bearing debt	\$ 1,250	\$ 5
All other liabilities	2,900	2,585
Shareholders' equity	188	11,861
Total liabilities and equity	\$ 4,338	\$14,451
Income Statement data:		
Total revenue	\$10,900	\$ 3,700
Interest expense	80	9
Net income	210	1,180

Requirements

- Before performing any calculations, which company do you think represents the better investment? Give your reason.
- Compute the EVA[®] for each company and then decide which company's shares you would rather hold as an investment. Assume both companies' cost of capital is 11.0% and use unadjusted net income in your calculations.

Challenge Exercises

E12-37. (*Learning Objectives 2, 3, 4: Using ratio data to reconstruct a company's Balance Sheet*) The following data (dollar amounts in millions) are taken from the financial statements of Floor 1 Industries, Inc.:

L0 2 3 4

Total liabilities	\$12,600
Pref. share	\$ 0
Total current assets	\$11,900
Accumulated depreciation	\$ 1,700
Debt ratio	70%
Current ratio	1.40

Requirement

- Complete the following condensed Balance Sheet. Report amounts to the nearest million dollars.

	(In millions)	
Current assets		
Property, plant and equipment		
Less: Accumulated depreciation		
Total assets		
Current liabilities		
Long-term liabilities		
Shareholders' equity		
Total liabilities and shareholders' equity		

L0 2 3 4

E12-38. (Learning Objectives 2, 3, 4: Using ratio data to reconstruct a company's Income Statement) The following data (dollar amounts in millions) are from the financial statements of County Corporation:

Average shareholders' equity.....	\$3,400
Interest expense.....	\$ 800
Preference shares.....	\$ 0
Operating income as a percent of sales.....	25%
Rate of return on shareholders' equity	12%
Income tax rate	40%

Requirement

1. Complete the following condensed Income Statement. Report amounts to the nearest million dollars.

Sales.....	
Operating expense.....	
Operating income.....	
Interest expense.....	
Pretax income	
Income tax expense	
Net income.....	

Quiz

Use the Fatima Bell Corporation financial statements that follow to answer questions 12–39 through 12–50.

A1				
	A	B	C	D
	Fatima Bell Corporation	(In millions, except per share amounts)		
	Consolidated Statements of Income			
1		Year ended December 31		
2		20X6	20X5	20X4
3				
4	Net revenue	\$ 42,810	\$ 35,299	\$ 30,968
5	Cost of goods sold	34,010	29,111	26,061
6	Gross profit	8,800	6,188	4,907
7	Operating expenses:			
8	Selling, general, and administrative	3,342	3,000	2,581
9	Research, development, and engineering	575	556	542
10	Special charges	—	—	502
11	Total operating expenses	3,917	3,556	3,625
12	Operating income	4,883	2,632	1,282
13	Investment and other income (loss), net	172	212	(78)
14	Income before income taxes	5,055	2,844	1,204
15	Income tax expense	1,108	912	472
16	Net income	\$ 3,947	\$ 1,932	\$ 732
17	Earnings per ordinary share:			
18	Basic	\$ 1.34	\$ 0.94	\$ 0.42
19				

A1			
	A	B	C
1	Fatima Bell Corporation		
2	Consolidated Statements of Financial Position		
3		December 31	
4	(In millions)	20X6	20X5
5	Assets		
6	Current assets:		
7	Cash and cash equivalents	\$ 4,301	\$ 4,138
8	Short-term investments	830	512
9	Accounts receivable, net	3,402	2,401
10	Inventories	427	410
11	Other	1,638	1,213
12	Total current assets	10,598	8,674
13	Property, plant and equipment, net	1,517	932
14	Investments	6,613	5,323
15	Other non-current assets	301	144
16	Total assets	\$ 19,029	\$ 15,073
17	Liabilities and Shareholders' Equity		
18	Current liabilities:		
19	Accounts payable	\$ 7,702	\$ 6,002
20	Accrued and other	3,676	3,044
21	Total current liabilities	11,378	9,046
22	Long-term debt	301	302
23	Other non-current liabilities	1,701	1,167
24	Commitments and contingent liabilities (Note 7)	—	—
25	Total liabilities	13,380	10,515
26	Shareholders' equity:		
27	Preference share and capital in excess of \$0.02		
28	par value; shares issued and outstanding: none	—	—
29	Ordinary share and capital in excess of \$0.05		
30	par value; shares authorized: 6,000; shares		
31	issued: 3,240 and 2,989, respectively	7,801	7,004
32	Treasury share, at cost; 175 and 124 shares,		
33	respectively	(6,333)	(4,404)
34	Retained earnings	4,321	2,054
35	Other comprehensive loss	(104)	(50)
36	Other	(36)	(46)
37	Total shareholders' equity	5,649	4,558
	Total liabilities and shareholders' equity	\$ 19,029	\$ 15,073

Q12-39. During 20X6, Fatima Bell's total assets

- a. increased by 26.2%.
- b. increased by \$9,390 million.
- c. increased by 20.8%.
- d. both a and b.

Q12-40. Fatima Bell's current ratio at year-end 20X6 is closest to

- a. 9,390.
- b. 1.2.
- c. 0.9.
- d. 20.8.

Q12-41. Fatima Bell's acid-test ratio at year-end 20X6 is closest to

- a. 0.75.
- b. 0.68.
- c. 0.45.
- d. \$8,533 million.

Q12-42. What is the largest single item included in Fatima Bell's debt ratio at December 31, 20X6?

- a. Investments
- b. Cash and cash equivalents
- c. Share capital
- d. Accounts payable

Q12-43. Using the earliest year available as the base year, the trend percentage for Fatima Bell's net revenue during 20X6 was

- a. 121%.
- b. up by 21.3%.
- c. up by \$11,842 million.
- d. 138%.

- Q12-44.** Fatima Bell's common-size Income Statement for 20X6 would report cost of goods sold as
 a. 130.5%. c. 79.5%.
 b. up by 16.8%. d. \$34,010 million.
- Q12-45.** Fatima Bell's cash conversion cycle during 20X6 was
 a. 102.7 days c. 91.7 days
 b. -44.3 days d. 100.9 days
- Q12-46.** Fatima Bell's receivable collection period during 20X6 was
 a. 29 days. c. 21 days.
 b. 25 days. d. 117 days.
- Q12-47.** Fatima Bell's long-term debt bears interest at 11%. During the year ended December 31, 20X6, Bell's times-interest-earned ratio was
 a. 137 times. c. 148 times.
 b. 144 times. d. 150 times.
- Q12-48.** Fatima Bell's trend of net profit margin is
 a. declining. c. stuck at 20.8%.
 b. worrisome. d. improving.
- Q12-49.** How many ordinary shares did Fatima Bell have outstanding, on average, during 20X6? Hint: Compute earnings per share.
 a. 2,946 million c. 5,244 million
 b. 5,258 million d. 2,965 million
- Q12-50.** The book value per share of Fatima Bell's ordinary share outstanding at December 31, 20X6, was
 a. \$5,649. c. \$1.96.
 b. \$1.84. d. \$2.08.

Problems MyLab Accounting

Select A and B problems can be found within MyLab Accounting, an online homework and practice environment. Your instructor may ask you to complete select problems using MyLab Accounting.

Group A

L0 1 4

P12-51A. (*Learning Objectives 1, 4: Computing trend percentages, return on sales, and comparison with the industry*) Net sales, net income, and total assets for Canberra Shipping Company for a five-year period follow:

(In thousands)	20X8	20X7	20X6	20X5	20X4
Net sales.....	\$672	\$518	\$378	\$330	\$320
Net income.....	39	35	51	38	30
Total liabilities	315	277	265	229	210

Requirements

1. Compute trend percentages for each item for 20X5 through 20X8. Use 20X4 as the base year and round to the nearest percent.
2. Compute the net profit margin for 20X6 through 20X8.
3. How does Canberra Shipping's net profit margin compare with that of the industry? In the shipping industry, rates above 5% are considered good, and rates above 7% are outstanding.

P12-52A. (*Learning Objectives 3, 4: Preparing common-size statements; analyzing profitability; making comparisons with the industry*) Top managers of McDonough Products, Inc., have asked for your help in comparing the company's profit performance and financial position with the average for the industry. The accountant has given you the company's Income Statement and Balance Sheet and also the following data for the industry:

L0 3 4

A1			
A			
McDonough Products, Inc. Income Statement Compared with Industry Average Year Ended December 31, 20X6			
1			
2		McDonough	Industry Average
3	Net sales	\$ 720,000	100.0%
4	Cost of goods sold	504,000	57.3
5	Gross profit	216,000	42.7
6	Operating expenses	180,000	29.4
7	Operating income	36,000	13.3
8	Other expenses	7,200	2.5
9	Net income	\$ 28,800	10.8%
10			

A1			
A			
McDonough Products, Inc. Balance Sheet Compared with Industry Average December 31, 20X6			
1			
2		McDonough	Industry Average
3	Current assets	\$ 486,400	72.1%
4	PPE, net	118,400	19.0
5	Intangible assets, net	21,760	4.8
6	Other assets	13,440	4.1
7	Total	\$ 640,000	100.0%
8			
9	Current liabilities	\$ 248,320	47.2%
10	Long-term liabilities	139,520	21.0
11	Shareholders' equity	252,160	31.8
12	Total	\$ 640,000	100.0%
13			

Requirements

1. Prepare a common-size Income Statement and Balance Sheet for McDonough Products. The first column of each statement should present McDonough Products' common-size statement, and the second column should show the industry averages.
2. For the profitability analysis, compute McDonough Products' (a) gross profit margin (b) operating profit margin, and (c) net profit margin. Compare these figures with the industry averages. Is McDonough Products' profit performance better or worse than the average for the industry?

3. For the analysis of financial position, compute McDonough Products' (a) current ratio, (b) quick ratio, and (c) debt ratio. Compare these ratios with the industry averages. Is McDonough Products' financial position better or worse than the average for the industry?

LO 4

■ writing assignment

P12-53A. (*Learning Objective 4: Calculating cash conversion cycle*). You are analyzing the effectiveness of the trading operations of AVN Limited. Extracts of its financial statements are provided below.

A1			
	A	B	C
1	AVN Limited		
2	Selected information from financial statements		
3		20X7	20X6
4	Sales	\$ 176,000	\$ 150,000
5	Cost of sales	83,000	73,000
6	Receivables, net	24,000	18,000
7	Inventory	16,000	12,000
8	Payables	20,000	16,000

Requirement

1. Calculate AVN's cash conversion cycle for 20X7. What is your assessment of AVN's cash conversion cycle?

LO 4

P12-54A. (*Learning Objective 4: Computing effects of business transactions on selected ratios*) Financial statement data of Greenland Engineering include the following items:

Cash	\$ 25,000	Accounts payable	\$101,000
Short-term investments.....	38,000	Accrued liabilities.....	37,000
Accounts receivable, net	82,000	Long-term notes payable.....	160,000
Inventories	149,000	Other long-term liabilities	37,000
Prepaid expenses	6,000	Net income.....	96,000
Total assets	674,000	Number of ordinary	
Short-term notes payable.....	41,000	shares outstanding	52,000

Requirements

- Compute Greenland's current ratio, debt ratio, and earnings per share. (Round all ratios to two decimal places.)
- Compute the three ratios after evaluating the effect of each transaction that follows. Consider each transaction *independently*.
 - Borrowed \$130,000 on a long-term note payable.
 - Issued 40,000 ordinary shares, receiving cash of \$400,000.
 - Paid short-term notes payable, \$25,000.
 - Purchased merchandise of \$48,000 on account, debiting Inventory.
 - Received cash on account, \$14,000.

P12-55A. (*Learning Objective 4: Using ratios to evaluate a share investment*) Comparative financial statement data of Bloomfield Optical Mart follow:

LO 4

A1			
	A	B	C
1	Bloomfield Optical Mart		
2	Comparative Income Statements		
3	Years Ended December 31, 20X6 and 20X5		
4		20X6	20X5
5	Net sales	\$ 695,000	\$ 595,000
6	Cost of goods sold	376,000	284,000
7	Gross profit	319,000	311,000
8	Operating expenses	127,000	142,000
9	Income from operations	192,000	169,000
10	Interest expense	37,000	51,000
11	Income before income tax	155,000	118,000
12	Income tax expense	40,000	53,000
13	Net income	\$ 115,000	\$ 65,000

A1				
	A	B	C	D
1	Bloomfield Optical Mart			
2	Comparative Balance Sheets			
3	December 31, 20X6 and 20X5			
4		20X6	20X5	20X4
5	Current assets:			
6	Cash	\$ 38,000	\$ 40,000	
7	Current receivables, net	217,000	149,000	\$ 140,000
8	Inventories	298,000	285,000	181,000
9	Prepaid expenses	9,000	25,000	
10	Total current assets	562,000	499,000	
11	Property, plant and equipment, net	284,000	276,000	
12	Total assets	\$ 846,000	\$ 775,000	710,000
13				
14	Total current liabilities	\$ 281,000	\$ 267,000	
15	Long-term liabilities	241,000	236,000	
16	Total liabilities	522,000	503,000	
17	Preference shareholders' equity, 5%, \$10 par	70,000	70,000	
18	Ordinary shareholders' equity, no par	254,000	202,000	195,000
19	Total liabilities and shareholders' equity	\$ 846,000	\$ 775,000	

Other information:

- Market price of Bloomfield ordinary share: \$82.20 at December 31, 20X6, and \$52.96 at December 31, 20X5.
- Ordinary shares outstanding: 20,000 during 20X6 and 18,000 during 20X5.

Requirements

- Compute the following ratios for 20X6 and 20X5:
 - Current ratio
 - Inventory turnover
 - Times-interest-earned ratio
 - Return on assets
 - Return on common shareholders' equity
 - Earnings per share
 - Price/earnings ratio

2. Decide whether (a) Bloomfield's financial position improved or deteriorated during 20X6 and (b) the investment attractiveness of Bloomfield's ordinary shares appears to have increased or decreased.
3. How will what you have learned in this problem help in the evaluation of an investment?

LO 4 5

■ writing assignment

P12-56A. (*Learning Objectives 4, 5: Using ratios to decide between two share investments; measuring economic value added*) Assume that you are considering purchasing shares as an investment. You have narrowed the choice to DVR.com and Express Shops and have assembled the following data.

Selected Income Statement data for the current year:

	DVR	Express
Net sales	\$605,000	\$520,000
Cost of goods sold	450,000	383,000
Income from operations	90,000	75,000
Interest expense.....	–	17,000
Net income.....	62,000	40,000

Selected Balance Sheet and market price data at end of current year:

	DVR	Express
Current assets:		
Cash	\$ 22,000	\$ 38,000
Short-term investments	10,000	14,000
Current receivables, net	182,000	167,000
Inventories	210,000	181,000
Prepaid expenses	21,000	8,000
Total current assets	445,000	408,000
Total assets	981,000	935,000
Total current liabilities	362,000	333,000
Total liabilities	673,000	700,000
Preference shares, 5%, \$150 par		30,000
Ordinary shares, \$1 par (100,000 shares)	100,000	
\$5 par (15,000 shares)		75,000
Total shareholders' equity	308,000	235,000
Market price per ordinary	\$ 6.10	\$ 55.00

Selected Balance Sheet data at *beginning* of current year:

	DVR	Express
Balance Sheet:		
Current receivables, net	\$144,000	\$195,000
Inventories	205,000	199,000
Total assets	853,000	908,000
Long-term debt	–	299,000
Preference share, 5%, \$150 par		30,000
Ordinary share, \$1 par (100,000 shares)	100,000	
\$5 par (15,000 shares)		75,000
Total shareholders' equity	260,000	221,000

Your strategy is to invest in companies that have low price/earnings ratios but appear to be in good shape financially. Assume that you have analyzed all other factors and that your decision depends on the results of ratio analysis.

Requirements

1. Compute the following ratios for both companies for the current year and decide which company's shares better fit your investment strategy.
 - a. Acid-test ratio
 - b. Inventory turnover
 - c. Receivables resident period
 - d. Debt ratio
 - e. Times-interest-earned ratio
 - f. Return on ordinary shareholders' equity
 - g. Earnings per ordinary share
 - h. Price/earnings ratio
2. Compute each company's economic-value-added (EVA[®]) measure and determine whether the companies' EVA[®]s confirm or alter your investment decision. Each company's cost of capital is 10%. Use unadjusted net income in your calculations.

Group B

P12-57B. (*Learning Objectives 1, 4: Computing trend percentages, return on sales equity, and comparison with the industry*) Net sales, net income, and total assets for Salvador Shipping Limited for a five-year period follow:

L0 1 4

(In thousands)	20X8	20X7	20X6	20X5	20X4
Net sales	\$656	\$538	\$381	\$330	\$320
Net income	37	33	50	38	30
Total liabilities	309	275	262	237	210

Requirements

1. Compute trend percentages for each item for 20X5 through 20X8. Use 20X4 as the base year and round to the nearest percent.
2. Compute the net profit margin for 20X6 through 20X8, rounding to three decimal places.
3. How does Salvador Shipping's net profit margin compare with that of the industry? In the shipping industry, rates above 5% are considered good, and rates above 7% are outstanding.

P12-58B. (*Learning Objectives 3, 4: Preparing common-size statements; analyzing portability; making comparisons with the industry*) Top managers of Walsh Products, Inc., have asked for your help in comparing the company's profit performance and financial position with the average for the industry. The accountant has given you the company's Income Statement and Balance Sheet and the following data for the industry:

L0 3 4

■ **writing assignment**

■ **spreadsheet**

A1			
	A	B	C
	Walsh Products, Inc.		
	Income Statement Compared with Industry Average		
	Year Ended December 31, 20X6		
1			
2		Walsh	Industry Average
3	Net sales	\$ 910,000	100.0%
4	Cost of goods sold	655,200	57.3
5	Gross profit	254,800	42.7
6	Operating expenses	218,400	29.4
7	Operating income	36,400	13.3
8	Other expenses	13,650	2.5
9	Net income	\$ 22,750	10.8%
10			

A1			
	A	B	C
1	Walsh Products, Inc.		
2	Balance Sheet Compared with Industry Average		
3	December 31, 20X6		
4		Walsh	Industry Average
5	Current assets	\$ 415,800	72.1%
6	PPE, net	101,520	19.0
7	Intangible assets, net	20,520	4.8
8	Other assets	2,160	4.1
9	Total	\$ 540,000	100.0%
10			
11	Current liabilities	\$ 209,520	47.2%
12	Long-term liabilities	114,480	21.0
13	Shareholders' equity	216,000	31.8
14	Total	\$ 540,000	100.0%
15			

Requirements

1. Prepare a common-size Income Statement and Balance Sheet for Walsh Products. The first column of each statement should present Walsh Products' common-size statement, and the second column should show the industry averages.
2. For the profitability analysis, compute Walsh Products' (a) gross profit margin, (b) operating profit margin, and (c) net profit margin. Compare these figures with the industry averages. Is Walsh Products' profit performance better or worse than the average for the industry?
3. For the analysis of financial position, compute Walsh Products' (a) current ratio, (b) quick ratio, and (c) debt ratio. Compare these ratios with the industry averages. Is Walsh Products' financial position better or worse than the average for the industry?

L0 4

P12-59B. (*Learning Objective 4: Calculating cash conversion cycle*) You are analyzing the effectiveness of the trading operations of CMI Limited. Extracts of its financial statements are provided below.

A1			
	A	B	C
1	CMI Limited		
2	Selected information from financial statements		
3		20X8	20X7
4	Sales	€ 1,545,000	€ 1,365,000
5	Cost of sales	855,000	730,000
6	Receivables, net	104,000	107,000
7	Inventory	89,000	75,000
8	Payables	93,000	69,000
9			

Requirement

1. Calculate CMI's cash conversion cycle for 20X8. What is your assessment of CMI's cash conversion cycle?

P12-60B. (*Learning Objective 4: Computing effects of business transactions on selected ratios*) Financial statement data of Tunisia Engineering include the following items:

LO 4

Cash	€ 26,000	Accounts payable	€106,000
Short-term investments.....	34,000	Accrued liabilities.....	34,000
Accounts receivable, net.....	87,000	Long-term notes payable.....	165,000
Inventories	145,000	Other long-term liabilities	32,000
Prepaid expenses	8,000	Net income.....	98,000
Total assets	677,000	Number of ordinary	47,000
Short-term notes payable.....	48,000	shares outstanding	€106,000

Requirements

1. Compute Tunisia's current ratio, debt ratio, and earnings per share.
2. Compute the three ratios after evaluating the effect of each transaction that follows. Consider each transaction *independently*.
 - a. Borrowed €110,000 on a long-term note payable.
 - b. Issued 20,000 shares of ordinary shares, receiving cash of €360,000.
 - c. Paid short-term notes payable, €28,000.
 - d. Purchased merchandise of €42,000 on account, debiting Inventory.
 - e. Received cash on account, €18,000.

P12-61B. (*Learning Objective 4: Using ratios to evaluate a share investment*) Comparative financial statement data of Schmid Optical Mart follow:

LO 4

■ **writing assignment**

A1			
	A	B	C
1	Schmid Optical Mart		
2	Comparative Income Statements		
3	Years Ended December 31, 20X6 and 20X5		
4		20X6	20X5
5	Net sales	\$ 680,000	\$ 585,000
6	Cost of goods sold	374,000	281,000
7	Gross profit	306,000	304,000
8	Operating expenses	130,000	143,000
9	Income from operations	176,000	161,000
10	Interest expense	30,000	49,000
11	Income before income tax	146,000	112,000
12	Income tax expense	40,000	45,000
13	Net income	\$ 106,000	\$ 67,000

A1				
	A	B	C	D
1	Schmid Optical Mart			
2	Comparative Balance Sheets			
3	December 31, 20X6 and 20X5			
4		20X6	20X5	20X4
5	Current assets:			
6	Cash	€ 32,000	€ 36,000	
7	Current receivables, net	211,000	154,000	€ 134,000
8	Inventories	291,000	288,000	188,000
9	Prepaid expenses	6,000	30,000	
10	Total current assets	540,000	508,000	
11	Property, plant and equipment, net	288,000	278,000	
12	Total assets	€ 828,000	€ 786,000	704,000
13	Total current liabilities	€ 280,000	€ 293,000	
14	Long-term liabilities	240,000	231,000	
15	Total liabilities	520,000	524,000	
16	Preference shareholders' equity, 3%, \$5 par	65,000	65,000	
17	Ordinary shareholders' equity, no par	243,000	197,000	198,000
18	Total liabilities and shareholders' equity	€ 828,000	€ 786,000	

Other information:

1. Market price of Schmid ordinary share: €78.12 at December 31, 20X6, and €59.10 at December 31, 20X5.
2. Ordinary shares outstanding: 19,000 during 20X6 and 17,000 during 20X5.

Requirements

1. Compute the following ratios for 20X6 and 20X5:
 - a. Current ratio
 - b. Inventory turnover
 - c. Times-interest-earned ratio
 - d. Return on ordinary shareholders' equity
 - e. Earnings per ordinary share
 - f. Price/earnings ratio
2. Decide whether (a) Schmid's financial position improved or deteriorated during 20X6 and (b) the investment attractiveness of Schmid's ordinary shares appears to have increased or decreased.
3. How will what you have learned in this problem help in the evaluation of an investment?

L0 4 5

■ writing assignment

P12-62B. (*Learning Objectives 4, 5: Using ratios to decide between two share investments; measuring economic value added*) Assume that you are considering purchasing shares as an investment. You have narrowed the choice to CDROM and E-shop Stores and have assembled the following data.

Selected Income Statement data for current year:

	CDROM	E-Shop
Net sales (all on credit).....	\$595,000	\$514,000
Cost of goods sold.....	453,000	386,000
Income from operations	88,000	69,000
Interest expense.....	—	12,000
Net income	66,000	37,000

Selected Balance Sheet and market price data at the *end* of the current year:

	CDROM	E-Shop
Current assets:		
Cash	€ 24,000	€ 41,000
Short-term investments	5,000	15,000
Current receivables, net	185,000	165,000
Inventories	219,000	187,000
Prepaid expenses	21,000	11,000
Total current assets	454,000	419,000
Total assets	978,000	928,000
Total current liabilities	363,000	332,000
Total liabilities	663,000	693,000
Preference shares, 6%, \$150 par		30,000
Ordinary shares, \$1 par (100,000 shares)	100,000	
\$5 par (10,000 shares)		50,000
Total shareholders' equity	315,000	235,000
Market price per ordinary share	€ 8.84	€ 70.68

Selected Balance Sheet data at the *beginning* of the current year:

	CDROM	E-Shop
Balance Sheet:		
Current receivables, net	€143,000	€190,000
Inventories	202,000	195,000
Total assets	843,000	914,000
Long-term debt	–	300,000
Preference share, 6%, €150 par		30,000
Ordinary share, €1 par (100,000 shares)	100,000	
€5 par (10,000 shares)		50,000
Total shareholders' equity	259,000	220,000

Your strategy is to invest in companies that have low price/earnings ratios, but which appear to be in good shape financially. Assume that you have analyzed all other factors and that your decision depends on the results of ratio analysis.

Requirements

- Compute the following ratios for both companies for the current year and decide which company's shares better fit your investment strategy.
 - Acid-test ratio
 - Inventory turnover
 - Receivables resident period
 - Debt ratio
 - Times-interest-earned ratio
 - Return on ordinary shareholders' equity
 - Earnings per share
 - Price/earnings ratio
- Compute each company's economic-value-added (EVA[®]) measure and determine whether the companies' EVA[®]s confirm or alter your investment decision. Each company's cost of capital is 12%. Use unadjusted net income in your calculations.

APPLY YOUR KNOWLEDGE

Decision Cases

L0 5

Case 1. (*Learning Objective 5: Assessing the effects of transactions on a company*) Suppose Nestlé is having a bad year in 20X4, as the company has incurred a \$4.9 billion net loss. The loss has pushed most of the return measures into the negative column and the current ratio dropped below 1.0. The company's debt ratio is still only 0.27. Assume top management of the company is pondering ways to improve the company's ratios. In particular, management is considering the following transactions:

1. Sell off the cable television segment of the business for \$30 million (receiving half in cash and half in the form of a long-term note receivable). The book value of the cable television business is \$27 million.
2. Borrow \$100 million on long-term debt.
3. Purchase treasury share for \$500 million cash.
4. Write off one-fourth of goodwill carried on the books at \$128 million.
5. Sell advertising at the normal gross profit of 60%. The advertisements run immediately.
6. Purchase trademarks from other companies, paying \$20 million cash and signing a one-year note payable for \$80 million.

Requirements

1. Top management wants to know the effects of these transactions (increase, decrease, or no effect) on the following ratios of Nestlé:
 - a. Current ratio
 - b. Debt ratio
 - c. Times-interest-earned ratio
 - d. Return on equity
 - e. Book value per ordinary share
2. Some of these transactions have an immediate positive effect on the company's financial condition. Some are definitely negative. Others have an effect that cannot be judged as clearly positive or negative. Evaluate each transaction's effect as positive, negative, or unclear. (Challenge)

■ writing assignment

L0 4

■ writing assignment

Case 2. (*Learning Objective 4: Analyzing the effects of an accounting difference on the ratios*) Assume that you are a financial analyst. You are trying to compare the financial statements of CNH Global, an international company that uses international financial reporting standards (IFRS), to those of Caterpillar, Inc., which uses US GAAP. Caterpillar, Inc., uses the last-in, first-out (LIFO) method to account for its inventories. IFRS does not permit CNH Global to use LIFO. Analyze the effect of this difference in accounting methods on the two companies' ratio values. For each ratio discussed in this chapter, indicate which company will have the higher (and the lower) ratio value. Also identify those ratios that are unaffected by the FIFO/LIFO difference. Ignore the effects of income taxes, and assume inventory costs are increasing. Then, based on your analysis of the ratios, summarize your conclusions as to which company looks better overall.

Case 3. (*Learning Objectives 2, 4: Identifying action to cut losses and establish profitability*) Suppose you manage Europe Vacations, a travel agency specializing in European tour destinations that lost money during the past year. To turn the business around, you must analyze the company and industry data for the current year to learn what is wrong. The company's data follow:

L0 2 4

A1			
	A	B	C
1	Europe Vacations, Inc.		
2	Common-Size Balance Sheet Data		
3		Outward Bound	Industry Average
4	Cash and short-term investments	3.0%	6.8%
5	Trade receivables, net	15.2	11.0
6	Inventory	64.2	60.5
7	Prepaid expenses	1.0	0.0
8	Total current assets	83.4%	78.3%
9	Fixed assets, net	12.6	15.2
10	Other assets	4.0	6.5
11	Total assets	100.0%	100.0%
12	Notes payable, short-term, 12%	17.1%	14.0%
13	Accounts payable	21.1	25.1
14	Accrued liabilities	7.8	7.9
15	Total current liabilities	46.0	47.0
16	Long-term debt, 11%	19.7	16.4
17	Total liabilities	65.7	63.4
18	Ord. shareholders' equity	34.3	36.6
19	Total liabilities and shareholders' equity	100.0%	100.0%
20			

A1			
	A	B	C
1	Europe Vacations, Inc.		
2	Common-Size Income Statement Data		
3		Outward Bound	Industry Average
4	Net sales	100.0%	100.0%
5	Cost of sales	(68.2)	(64.8)
6	Gross profit	31.8	35.2
7	Operating expense	(37.1)	(32.3)
8	Operating income (loss)	(5.3)	2.9
9	Interest expense	(5.8)	(1.3)
10	Other revenue	1.1	0.3
11	Income (loss) before income tax	(10.0)	1.9
12	Income tax (expense) saving	4.4	(0.8)
13	Net income (loss)	(5.6)%	1.1%

Requirement

- On the basis of your analysis of these figures, suggest four courses of action Europe Vacations might take to reduce its losses and establish profitable operations. Give your reason for each suggestion. (Challenge)



Ethical Issue

Turnberry Golf Corporation's long-term debt agreements make certain demands on the business. For example, Turnberry may not purchase treasury shares in excess of the balance of retained earnings. Also, long-term debt may not exceed shareholders' equity, and the current ratio may not fall below 1.50. If Turnberry fails to meet any of these requirements, the company's lenders have the authority to take over management of the company.

Changes in consumer demand have made it hard for Turnberry to attract customers. Current liabilities have mounted faster than current assets, causing the current ratio to fall to 1.47. Before releasing financial statements, Turnberry's management is scrambling to improve the current ratio. The controller points out that the company owns an investment that is currently classified as long-term. The investment can be classified as either long-term or short-term, depending on management's intention. By deciding to convert an investment to cash within one year, Turnberry can classify the investment as short-term—a current asset. On the controller's recommendation, Turnberry's board of directors votes to reclassify long-term investments as short-term.

Requirements

1. What is the accounting issue in this case? What ethical decision needs to be made?
2. Who are the stakeholders?
3. Analyze the potential impact on the stakeholders from the following standpoints: (a) economic, (b) legal, and (c) ethical.
4. Shortly after the financial statements are released, sales improve; so, too, does the current ratio. As a result, Turnberry's management decides not to sell the investments it had reclassified as short term. Accordingly, the company reclassifies the investments as long term. Has management acted unethically? Give the reasoning underlying your answer.

■ writing assignment

■ writing assignment

Focus on Financials: | Nestlé

Refer to Nestlé's consolidated financial statements in Appendix A (available on www.pearson-globaleditions.com/Harrison).

Requirements

Use the consolidated financial statements and the data in Nestlé's annual report (Appendix A) to evaluate the company's comparative performance for 2016 versus 2015.

1. Does Nestlé appear to be improving or declining in the following dimensions?
 - a. The ability to pay its current liabilities
 - b. The ability to sell inventory and collect receivables
 - c. The ability to pay long-term debts
 - d. Profitability
 - e. The ability to generate cash flows
 - f. The potential of the company's shares as a long-term investment. You may want to obtain Nestlé's current share price to support your answer. (Challenge)
2. What is your opinion of the company's outlook for the future? Would you buy the company's share as an investment? Why or why not? (Challenge)

■ writing assignment

Group Projects

■ writing assignment

Project 1. Select an industry you are interested in, and use the leading company in that industry as the benchmark. Then select two other companies in the same industry. For each category of ratios in the Decision Guidelines featured earlier in this chapter, compute at least two ratios for all three companies. Write a two-page report that compares the two companies with the benchmark company.

Project 2. Select a company in the retail industry and obtain its financial statements. Convert the Income Statement and the Balance Sheet to common size and compare the company you selected to the industry averages. You can assume the home furnishings' industry averages you have seen in this chapter are representative of the retail industry as a whole.

Quick Check Answers

1. *a* ($\$1,822/\$1,731$)
2. *c* ($\$1,822/\$17,710$)
3. *c* ($\$2,694 - \$1,128)/\$1,128 = 138.8\%$)
4. *d* ($\$1,780/\$26,060$)
5. *b*
6. *b* ($\{3,302/[(140 + 120)/2]\} = 25.4 \approx 25$ times)
7. *d* ($[(\$690 + \$780)/\$2,374 = 0.62]$)
8. *a* ($(\$780 + \$840)/2 / (\$17,710 / 365) = 16.7 \approx 17$ days)
9. *b* (Debt ratio is $(\$26,060 - \$13,291)/\$26,060 = 0.49$. This debt ratio is lower than the average for most companies, given in the chapter as 0.64.)
10. *d* ($\$2,694/\$17,710 = 0.152$)
11. *c* ($\$2,694/ [(\$13,291 + \$10,615) / 2] = 0.225$)
12. *d* ($\$30/\2.69)
13. *d* ($\$0.50/\30)
14. *a* ($[\$2,694 + \$372 - (\$305 + \$9,500 + \$10,615) \times 0.05] = \$2,045$)